

COSE471: Data Science

Classification (1)

Department of Computer Science and Engineering

Korea University

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April 24, 2026

Course Schedule

Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1		<i>Latex, Preprocessing/Pattern mining</i>
5	Apr 03	Clustering		
6	Apr 10	Outlier Detection & Invited Talk		<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	● HW (1&2 merged)	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification	●	
9	May 01	Classification	●	
10	May 08	Tutorial 3	●	<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	● Project	
13	May 29	Project discussion	●	
14	Jun 05	Project discussion	●	
15	Jun 12	Final presentation	●	
16	Jun 19	Wrap-up and Final report	●	

Announcement

❑ Homework is released!

- You can submit your responses through Google Form
 - Part 1: Data and Pattern Mining
 - Part 2: Clustering
 - Part 3: Outlier Detection and Classification
- Due on May 8 11:59PM
 - Scores and answers will be out on May 12

❑ Hybrid class on May 1 (holiday)

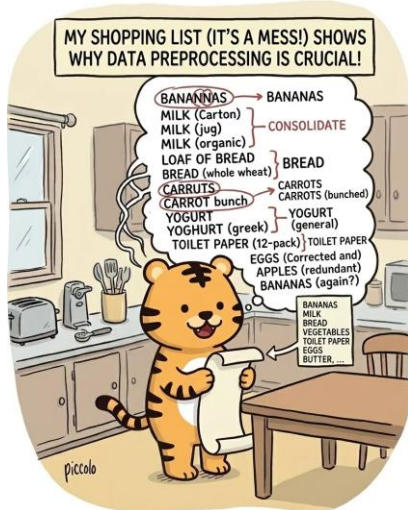
- Offline and Online (via Zoom, will be recorded and uploaded)

Last Lecture

- ❑ Basic concepts of outlier detection
 - Motivation
 - Comparison with other tasks
 - Types, Challenges, and Approaches
- ❑ Statistical methods
 - Parametric methods
 - Non-parametric methods
- ❑ Proximity-based methods
 - Distance-based (kNN)
 - Density-based (LOF, KDE)
- ❑ Methods for high-dimensional data
 - Proximity-based
 - Angle-based
 - Ensemble-based
- ❑ Outlier detection evaluation



RECAP



Outline

- ❑ **Basic concepts of classification**
- ❑ Classification methodologies
- ❑ Classification evaluation
- ❑ Ensemble classifiers



Classification

□ Dataset

- A set of data instances x with attributes and class label y , where
 - Attributes (a.k.a. feature, predictor, independent variable, input)
 - e.g., $x = (\text{Gender}, \text{Name}, \text{Grade}, \text{Body Temp}, \text{Age})$
 - Class label (a.k.a. response, dependent variable, output)
 - e.g., $y = \text{Covid}$

ID	Gender	COVID	Name	Grade	Body Temp	Age	...
1	Male	Positive	Kim	Junior	38.5	24	...
2	Female	Negative	Sarah	Graduated	36.0	35	...
...			...	...	.	...	...
100	Female	Negative	Park	Sophomore	36.5	22	...

Classification

□ Dataset

- A set of data instances x with attributes and class label y , where
 - Attributes (a.k.a. feature, predictor, independent variable, input)
 - Class label (a.k.a. class, label, response, dependent variable, output)

□ Classification task

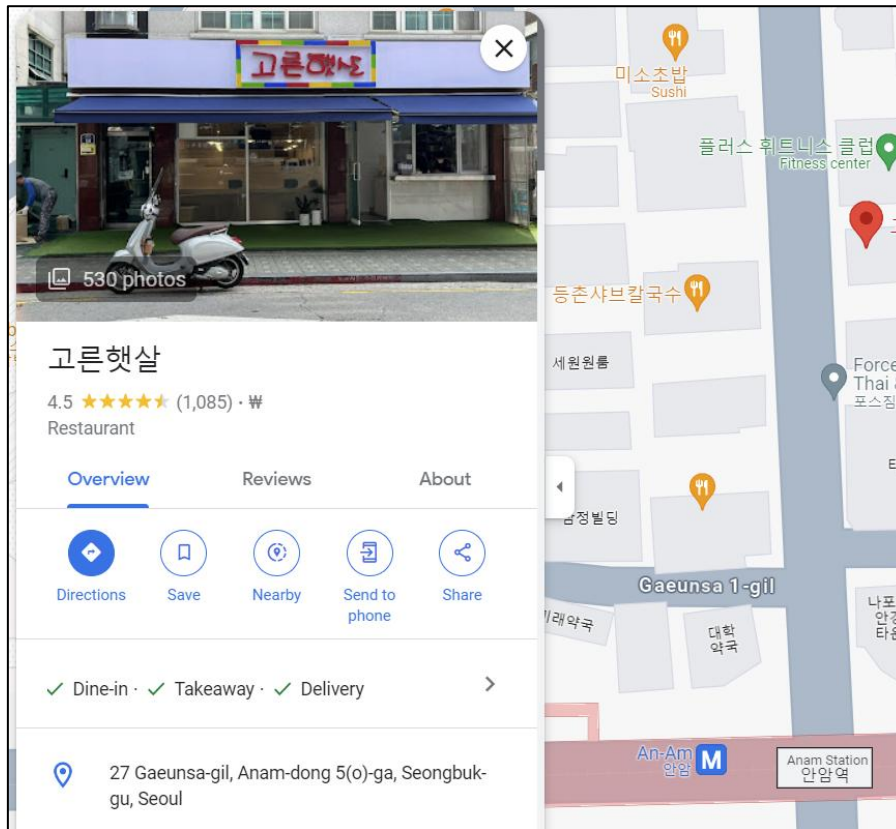
- Learn a model that maps a set of attributes x into one of the predefined class labels y
 - e.g., for a student 1 (Gender = Male, Name = Kim, Grade = Junior, BodyTemp = 38.5, Age = 24), COVID = Pos or Neg?

ID	Gender	COVID	Name	Grade	Body Temp	Age	...
1	Male	Positive	Kim	Junior	38.5	24	...
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...			...	...	.	...	...
100	Female	Negative	Park	Sophomore	36.5	22	...

Classification > Examples

□ *What can be attributes and class labels for this restaurant data?*

- *i.e., what information do we want to use to predict what?*



★★★★★ 4 months ago
It comes in a generous amount~
Tuna kimbap is very delicious.
There is a lot of rice, so it tastes good even if two people eat it.

Service
Dine in

Meal type
Dinner

Price per person
₩1-10,000

Food: 4

Service: 4

Atmosphere: 4

Recommended dishes
토종순대 참치김밥



Classification > Examples

❑ *What can be attributes and class labels for this restaurant data?*

고른햇살
4.5 ★★★★★ (1,085) · ₩ Restaurant

Overview Reviews About

Directions Save Nearby Send to phone Share

✓ Dine-in · ✓ Takeaway · ✓ Delivery

27 Gaeunsa-gil, Anam-dong 5(o)-ga, Seongbuk-gu, Seoul

★★★★★ 4 months ago
It comes in a generous amount~
Tuna kimbap is very delicious.
There is a lot of rice, so it tastes good even if two people eat it.

Service
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Atmosphere: 4

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토종순대 참치김밥

★★★★★ 8 months ago
The kimbap here is my love. Even if I eat it a thousand times, I never get tired of it! Tuna kimbap, cheese kimbap, and pork kimbap are all delicious!!
Not only does it taste good, it's also good value for money!

Classification > Examples

□ Real world applications

Task	Attribute set	Class labels
Customer Churn Prediction	Customer name, age, salary, purchase history, usage patterns, customer service interactions, etc.	Churn, Retain
Review Sentiment Analysis	Review product, image, keyword frequency, n-gram keywords, sentiment scores of keywords, etc.	Positive, Negative, Neutral
Restaurant Categorization	Interior, exterior, review, location, price, etc.	Asian, American, African, European, Middle Eastern, Latin American
Email Categorization	Sender's email address, subject line, email body text, time sent, frequency of emails from the same sender, etc.	Spam, Promotional, Personal, Work-related
Medical Diagnosis	Medical images, patient demographics, symptom descriptions, lab results, etc.	Positive, Negative
Predictive Maintenance	Sensor readings, machine operational data, error logs, maintenance history, etc.	Normal Operation, Maintenance Required, Urgent Repair

Outline

- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - **Lazy learning vs. eager learning**
 - k-nearest neighbors, Logistic regression, Support vector machine
 - Decision tree
- ❑ Classification evaluation
- ❑ Ensemble classifiers

Classification > Methodologies > Lazy vs. Eager



What breed of dog am I?

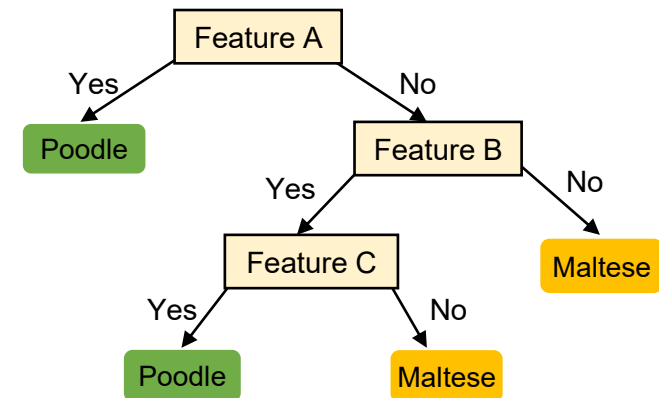
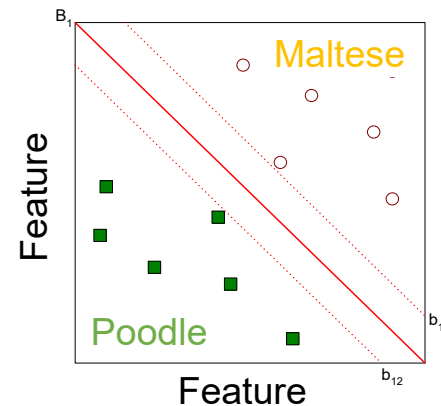
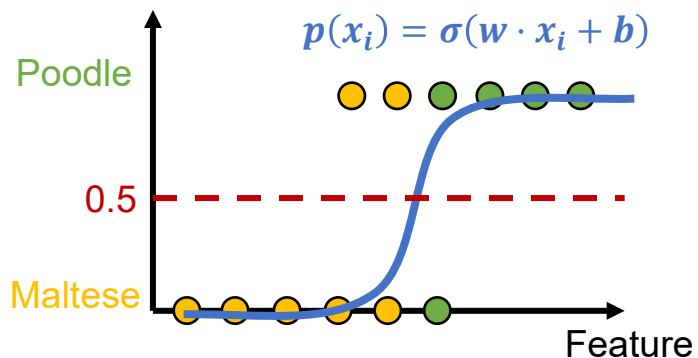
Classification > Methodologies > Lazy vs. Eager



Classification > Methodologies > Lazy vs. Eager

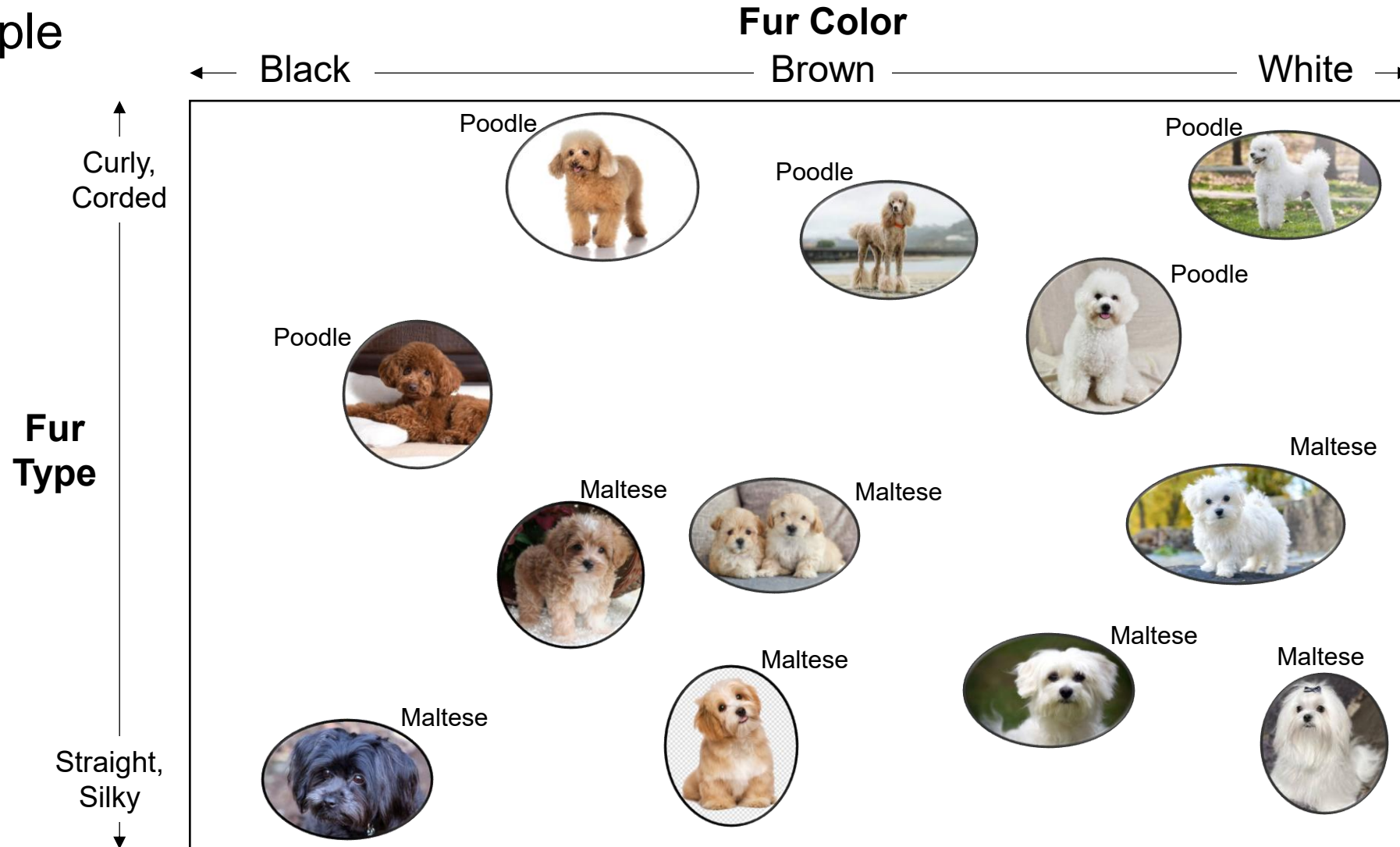
□ Philosophy

- Lazy learning: stores training data and classifies new data by referencing them directly
 - a.k.a., instance-based learning
 - e.g., **k-nearest neighbors**, locally weighted regression, case-based reasoning
- Eager learning: constructs a model from training data and classifies new data by the model
 - a.k.a., model-based learning
 - e.g., **Logistic regression, support vector machine, decision tree**



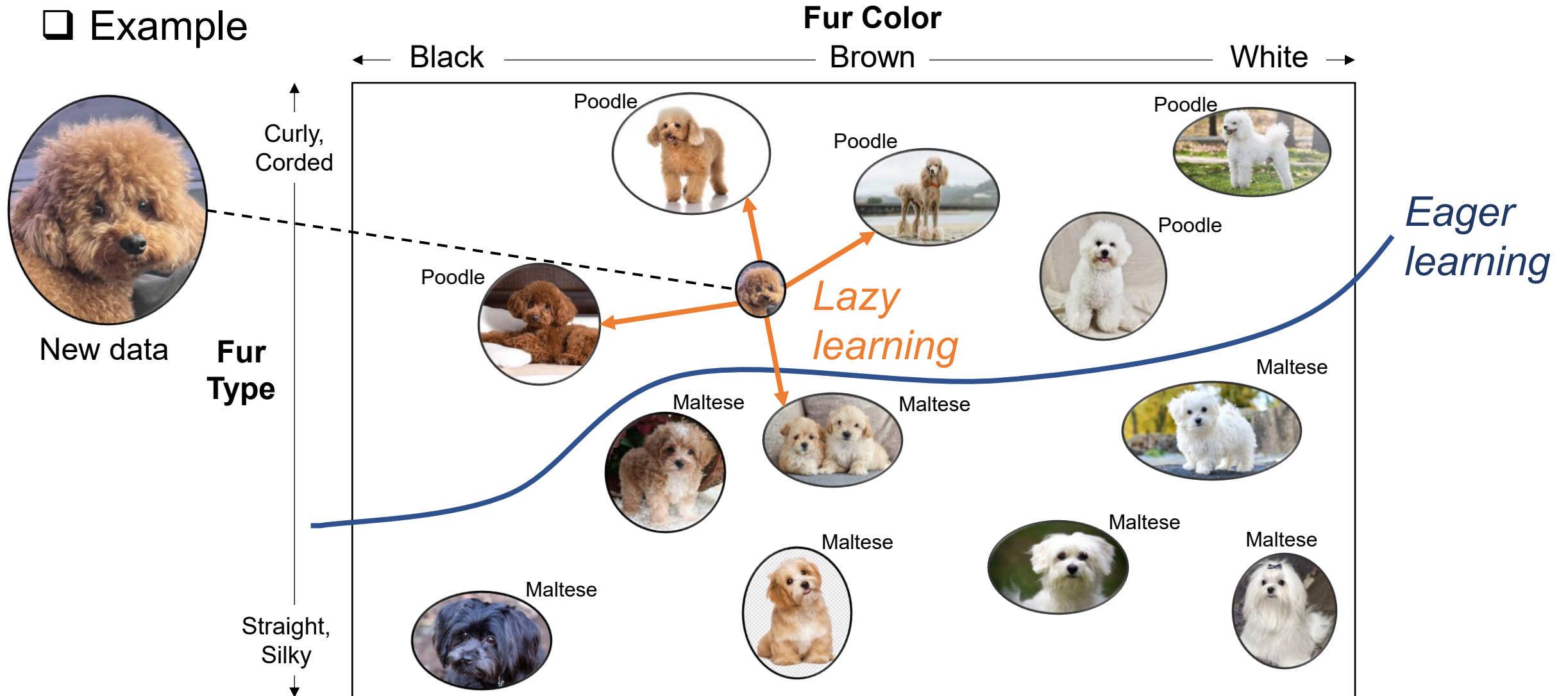
Classification > Methodologies > Lazy vs. Eager

□ Example



Classification > Methodologies > Lazy vs. Eager

Example



■ Classification > Methodologies > Lazy vs. Eager

□ Characteristics

▪ Lazy learning

- Use many local functions to form an implicit global approximation
- Flexible and adaptive to training data, robust to noise but potential for poor generalization
- Spend less time in training but more time in predicting

▪ Eager learning

- Form an explicit global approximation that covers the entire instance space
- Good generalization of training data but potential for high sensitivity to irrelevant features
- Spend more time in training but less time in predicting

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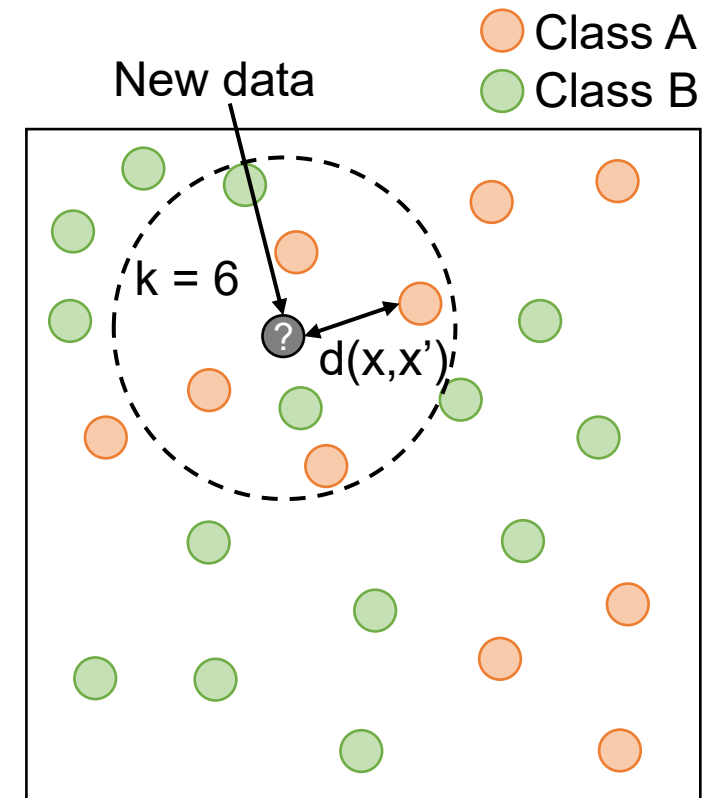
Classification > Methodologies > kNN

□ Main idea

- Given a new data, find the most similar data in the training dataset, and consult them to make a decision

□ Requirements

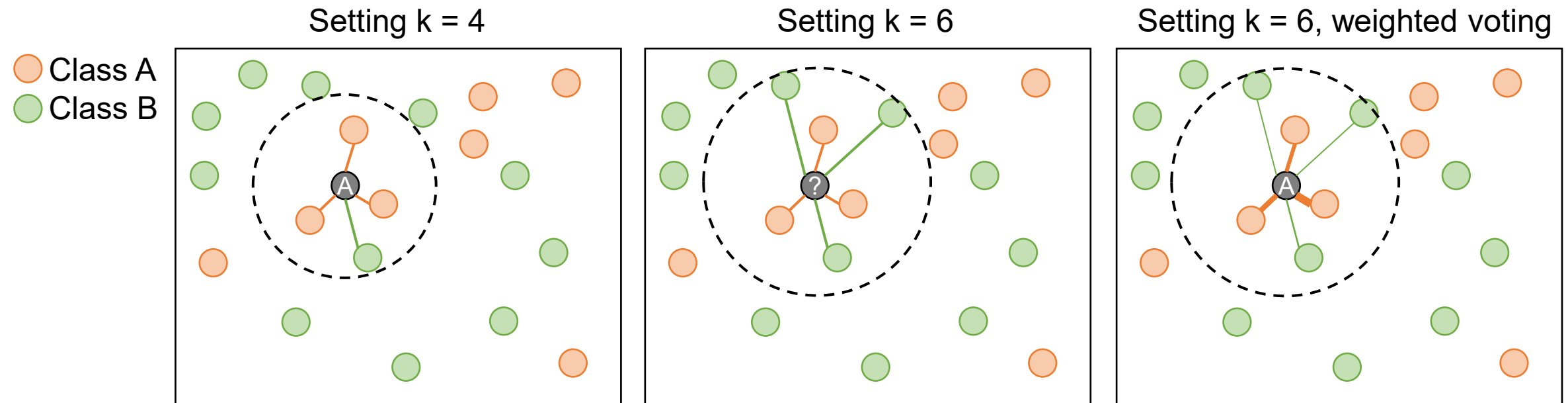
- A set of labeled data
- Proximity metric $d(x, x')$: distance/similarity between a pair of data instances x and x'
 - e.g., Euclidean distance, cosine distance, etc.
- The value of k : the number of nearest neighbors to retrieve
- Decision policy: a method for using k nearest neighbors of a target data x to determine a label for x
 - e.g., majority vote



Classification > Methodologies > kNN

□ Common procedure for classification [[demo](#)]

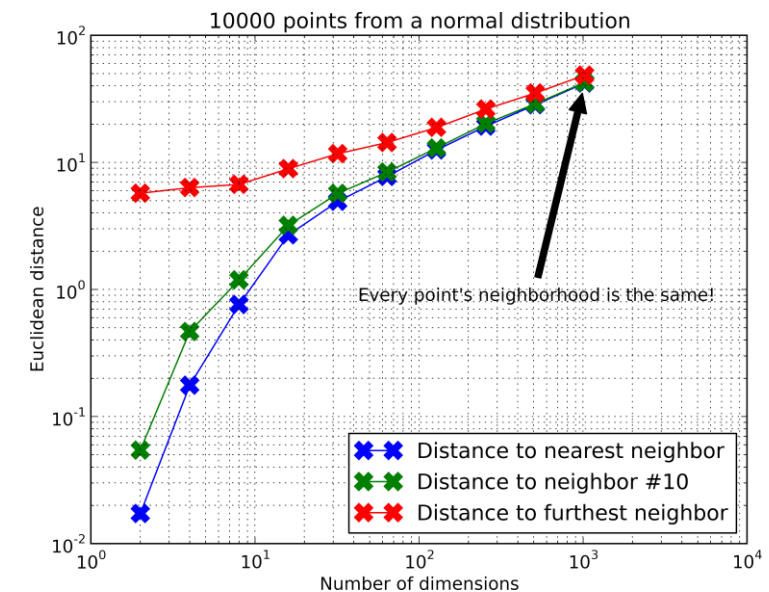
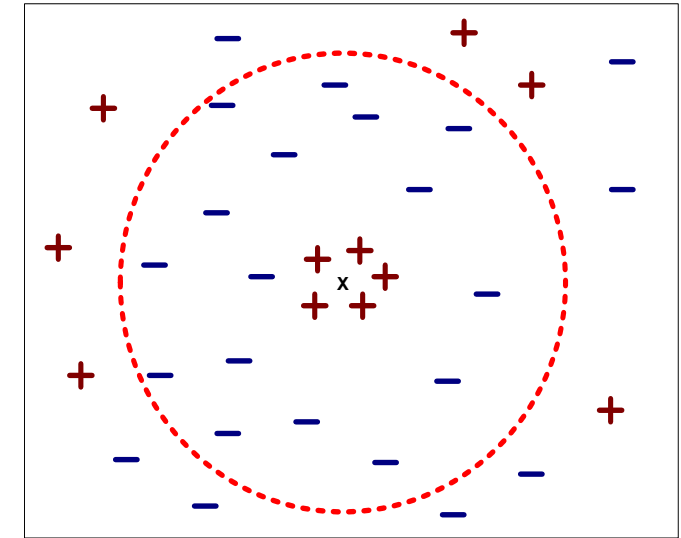
- Find the k-nearest neighbors of a new data x_q
 - The data points that have the k smallest distance to x_q
- Take the majority vote of class labels among the k-nearest neighbors
 - Can also weigh the vote according to distance (e.g., $w_i = \frac{1}{d(x_q, x_i)^2}$)



Classification > Methodologies > kNN

❑ Issues

- The value of k
 - Too small $k \rightarrow$ classification is sensitive to noise points (i.e., overfitting)
 - Too large $k \rightarrow$ may consider points from other classes (i.e., underfitting)
 - Rule of thumb is the square root of the number of samples (i.e., $k = \sqrt{n}$)
- Scaling
 - Attributes may have to be scaled to prevent distance measures from being dominated by one of the attributes (i.e., normalization)
 - e.g.,
 - The height of a person may vary from 1.5m to 1.8m
 - The weight of a person may vary from 40kg to 100kg
 - The income of a person may vary from \$10K to \$1M
- High-dimensional data
 - Curse of dimensionality: the distance becomes meaningless
 - Distance between neighbors could be dominated by irrelevant attributes



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Classification > Methodologies > Logistic Regression

□ [Review] Linear regression

▪ Goal

- Estimating the relationships between independent attributes (x) to continuous dependent value (y)
- e.g., {weight, age} → {tall}

▪ Data

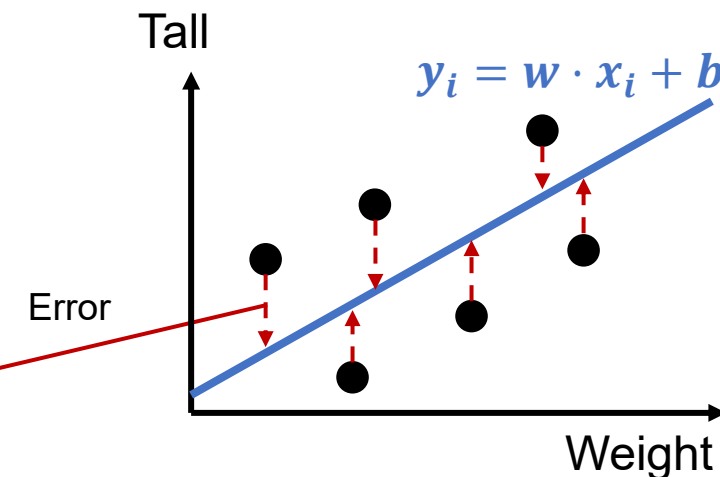
- n objects with p attributes $x_i = (x_{i1}, x_{i2}, \dots, x_{ip})$, $i = 1, 2, 3, \dots, n$
- Target values to predict: $y_i, i = 1, 2, 3, \dots, n$

▪ Model

- The weight vector w and bias b are the parameters learned from the data
- $y_i = w \cdot x_i + b$ where weight vector $w = (w_1, w_2, \dots, w_p)$

▪ Optimization

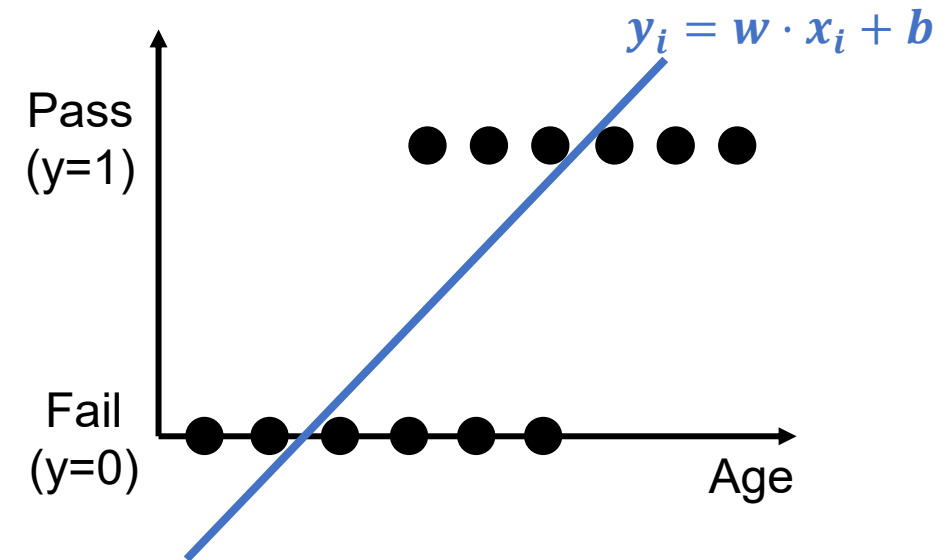
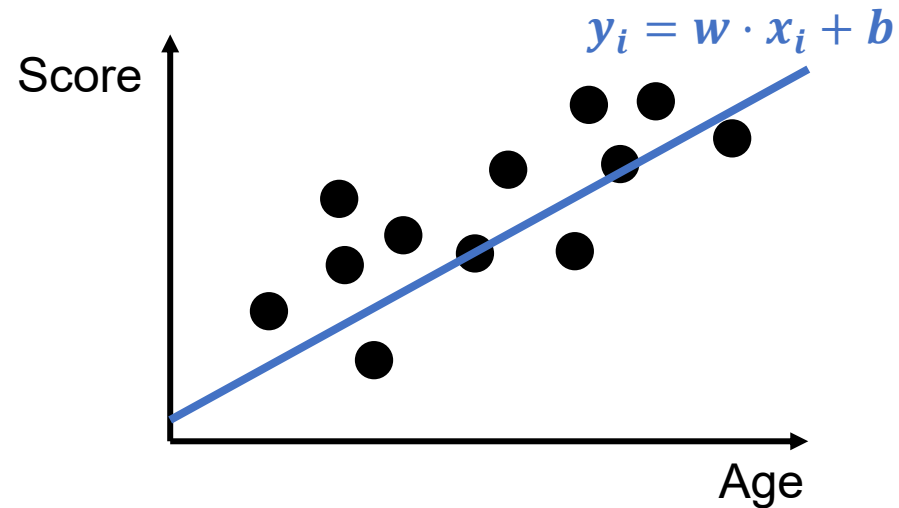
- Minimize the least squares error (LSE): $L(w, b) = \sum_{i=1}^n (y_i - w \cdot x_i - b)^2$
 - e.g., Closed-form solution or gradient descent



Classification > Methodologies > Logistic Regression

□ How to solve “classification” problem by regression?

- e.g., predicting scores by age → classifying age into Pass or Fail in the test



Classification > Methodologies > Logistic Regression

❑ Key idea: transform the real value y_i into a probability value in $[0,1]$

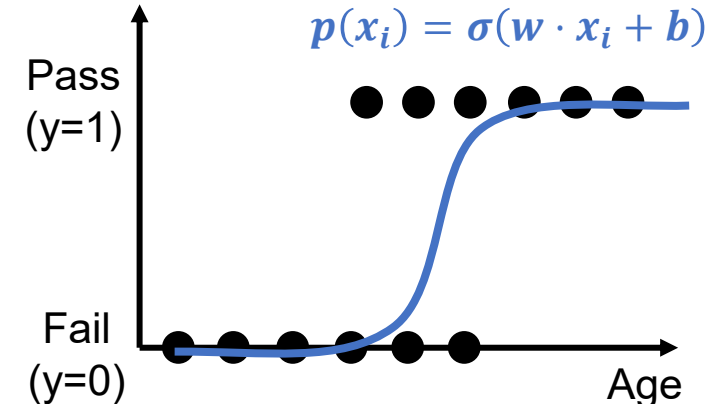
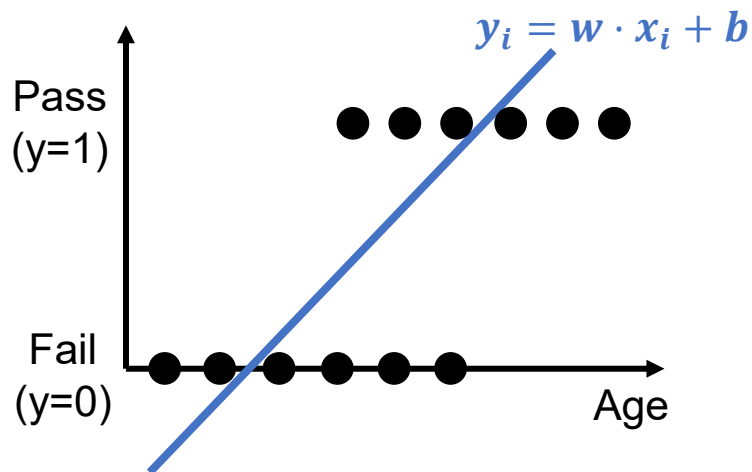
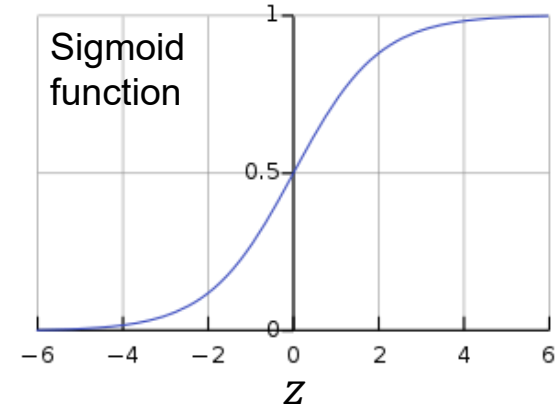
- Sigmoid function (a.k.a. logistic function)

- $\sigma(z) = \frac{1}{1+e^{-z}} \rightarrow$ projects $z \sim (-\infty, +\infty)$ to $[0, 1]$, differentiable

- Pass $y_i = w \cdot x_i + b$ into the sigmoid function to get a probability in $[0,1]$

- $p(Y = 1|x_i; w, b) = p(x_i) = \sigma(w \cdot x_i + b)$

- $p(Y = 0|x_i; w, b) = 1 - p(x_i) = 1 - \sigma(w \cdot x_i + b)$



Classification > Methodologies > Logistic Regression

□ Key idea: transform the real value y_i into a probability value in $[0,1]$

■ Model

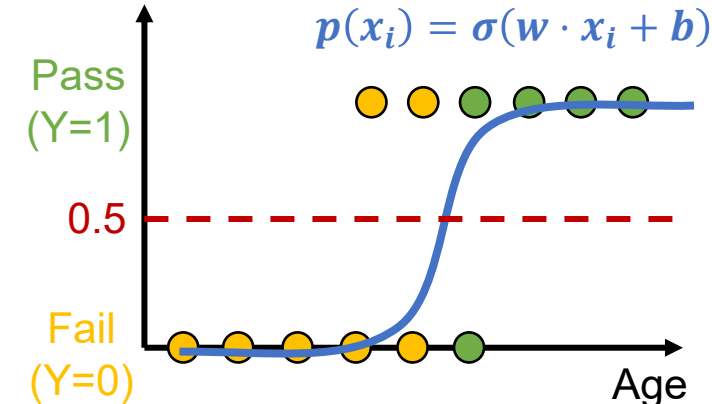
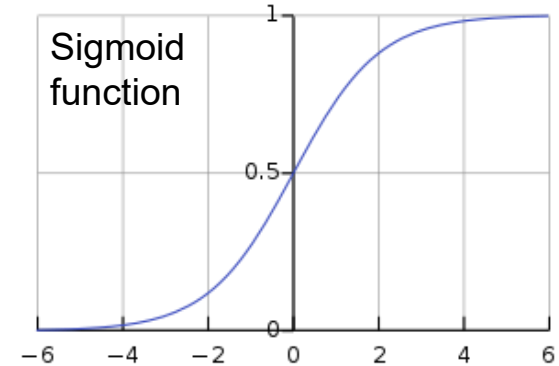
- $p(Y = 1|X = x_i; w, b) = p(x_i) = \sigma(w \cdot x_i + b)$
- $p(Y = 0|X = x_i; w, b) = 1 - p(x_i) = 1 - \sigma(w \cdot x_i + b)$
- $p(Y = y_i|X = x_i; w, b) = p(x_i)^{y_i}(1 - p(x_i))^{1-y_i}$

■ Optimization

- Maximum likelihood estimation (MLE)
 - $\prod_{i=1}^N p(Y = y_i|X = x_i; w, b) = \prod_{i=1}^N p(x_i)^{y_i}(1 - p(x_i))^{1-y_i}$
- Minimize the negative log likelihood loss (a.k.a. cross-entropy loss)
 - $L(w, b) = -\sum_{i=1}^n (y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i)))$

■ Interpretation

- if $p(x_i) = \sigma(w \cdot x_i + b) > 0.5$, $Y = 1$, otherwise $Y = 0$
 - The threshold can be adjusted if needed

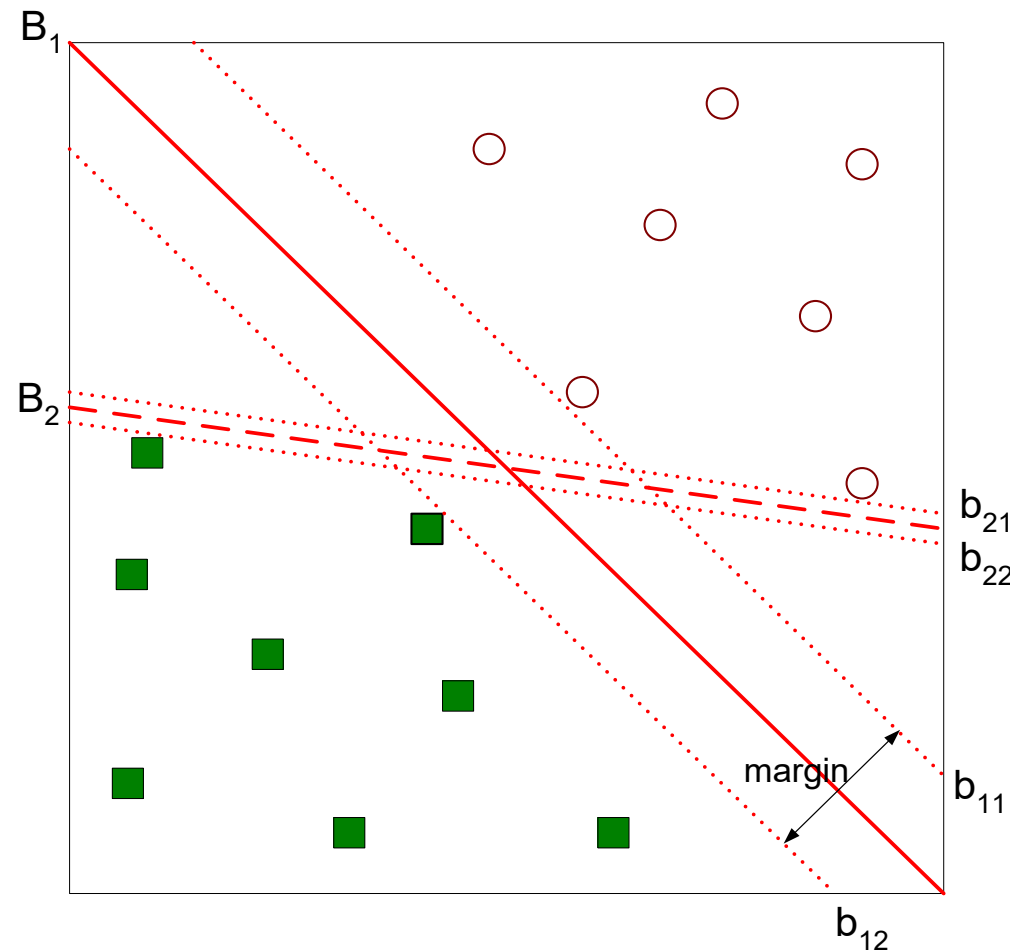


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- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - Lazy learning vs. eager learning
 - k-nearest neighbors, Logistic regression, **Support vector machine**
 - Decision tree
- ❑ Classification evaluation
- ❑ Ensemble classifiers

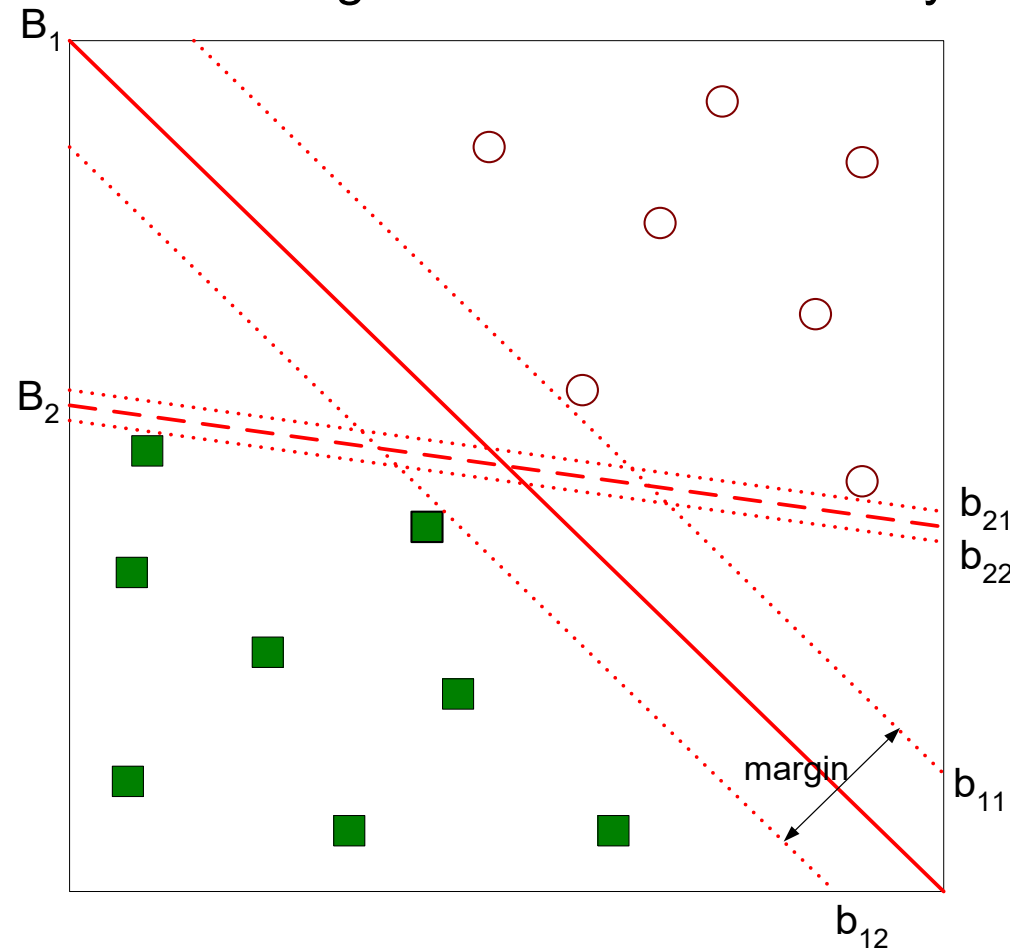
Classification > Methodologies > SVM

- Find a hyperplane (decision boundary) that separates the data



Classification > Methodologies > SVM

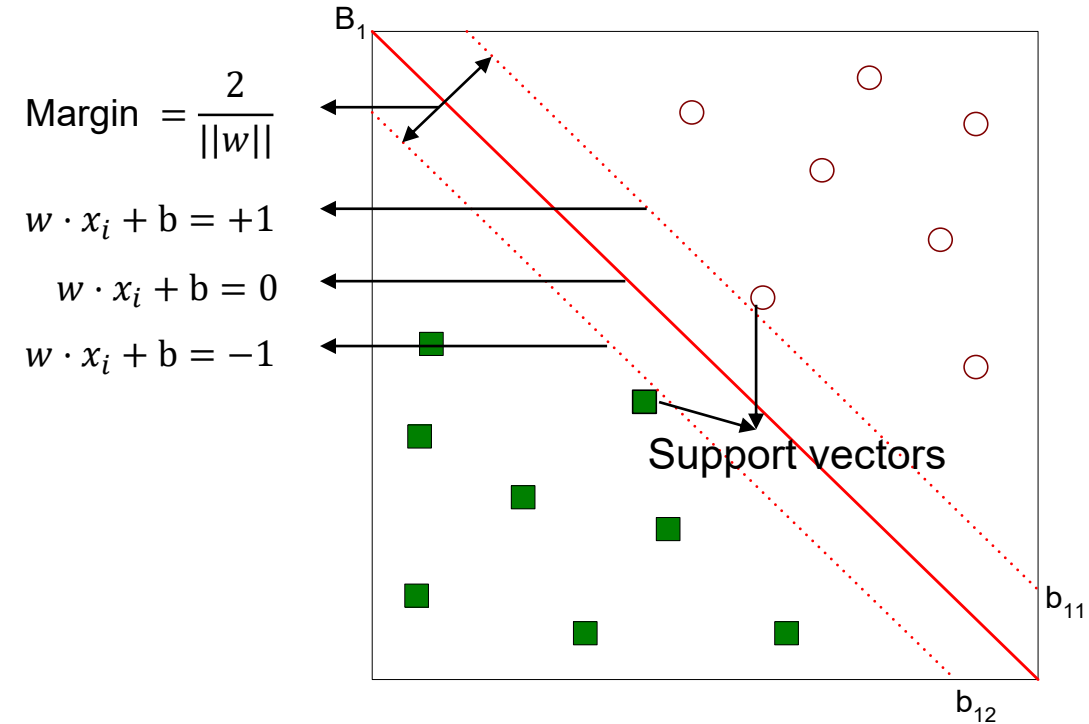
- Find a hyperplane (decision boundary) that separates the data
 - Which hyperplane maximizes the margin between the boundary instances? → B1 is better than B2



Classification > Methodologies > SVM

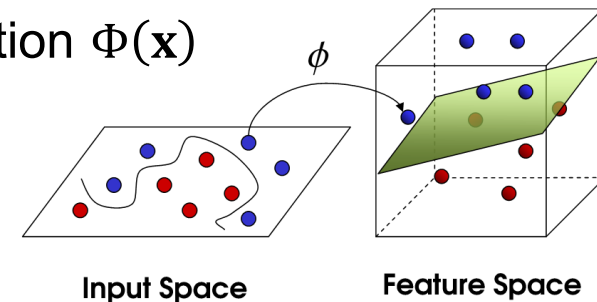
□ Linear SVM

- Find the hyperplane (w) that satisfies
 - Minimize $L(w) = \frac{\|w\|^2}{2}$
 - Subject to $y_i = \begin{cases} 1 & \text{if } w \cdot x_i + b \geq 1 \\ -1 & \text{if } w \cdot x_i + b \leq -1 \end{cases}$
- Extension: allowing small errors ξ_i (slacks)
 - Minimize $L(w) = \frac{\|w\|^2}{2} + C \sum_{i=1}^N \xi_i$
 - Subject to $y_i = \begin{cases} 1 & \text{if } w \cdot x_i + b \geq 1 - \xi_i \\ -1 & \text{if } w \cdot x_i + b \leq -1 + \xi_i \end{cases}$



□ Nonlinear SVM

- Find the hyperplane in a transformed space by a mapping function $\Phi(\mathbf{x})$
- Kernel trick: $K(\mathbf{x}_i, \mathbf{x}_j) = \Phi(\mathbf{x}_i) \cdot \Phi(\mathbf{x}_j)$
 - Polynomial $(1 + \mathbf{x}_i \cdot \mathbf{x}_j)^p$, Gaussian $(e^{-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}})$, ...



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Outline

- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - **Decision tree**
 - **Overview**
 - GINI index → CART
 - Entropy → ID3
 - Gain ratio → C4.5
- ❑ Classification evaluation
- ❑ Ensemble classifiers
- ❑ Summary

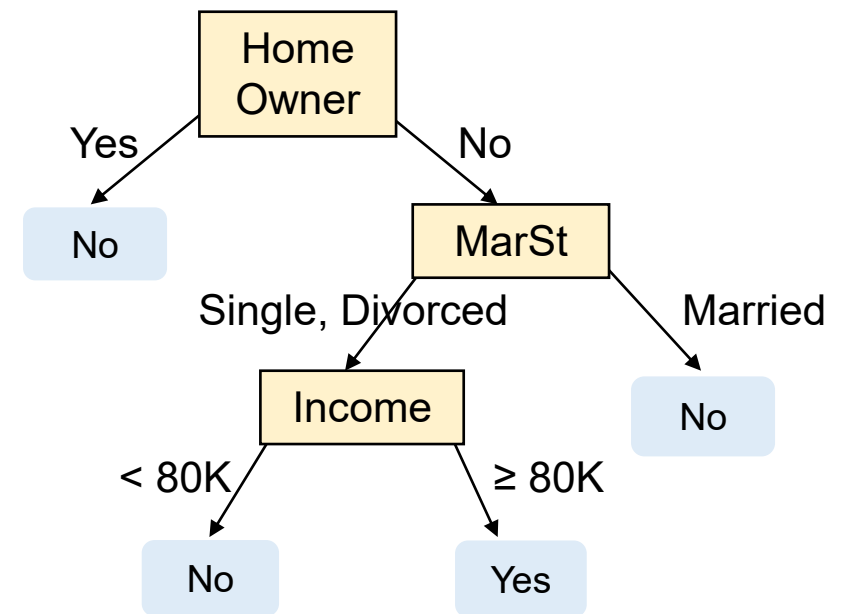
Classification > Methodologies > Decision Tree

- A top-down, recursive, divide-and-conquer process to classify data into labels

	Binary	Nominal	Ordinal	Numerical	Label
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

Tree
induction
➔
Learning
a model

Example decision tree



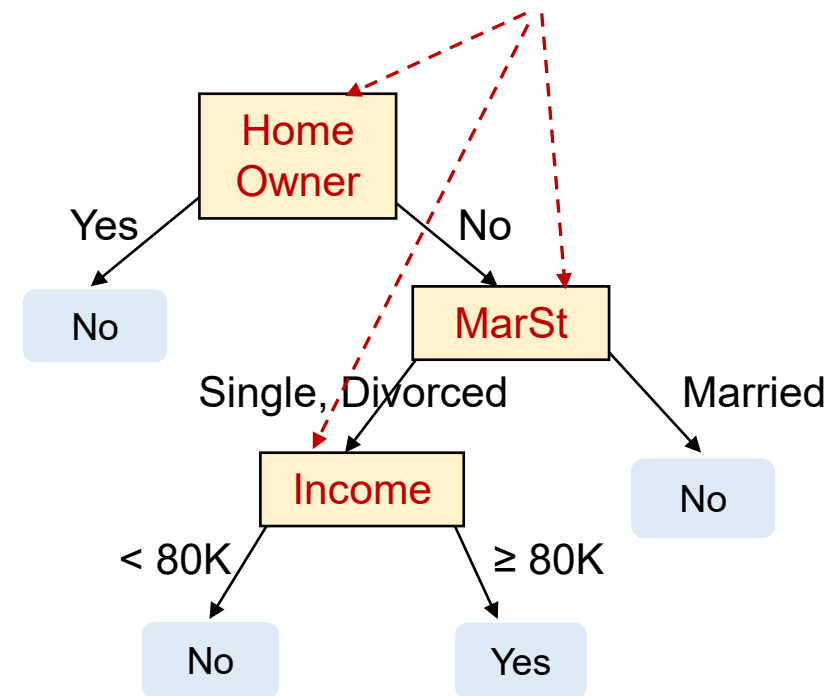
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Tree
induction
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Learning
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Splitting attributes (i.e., splitting nodes)

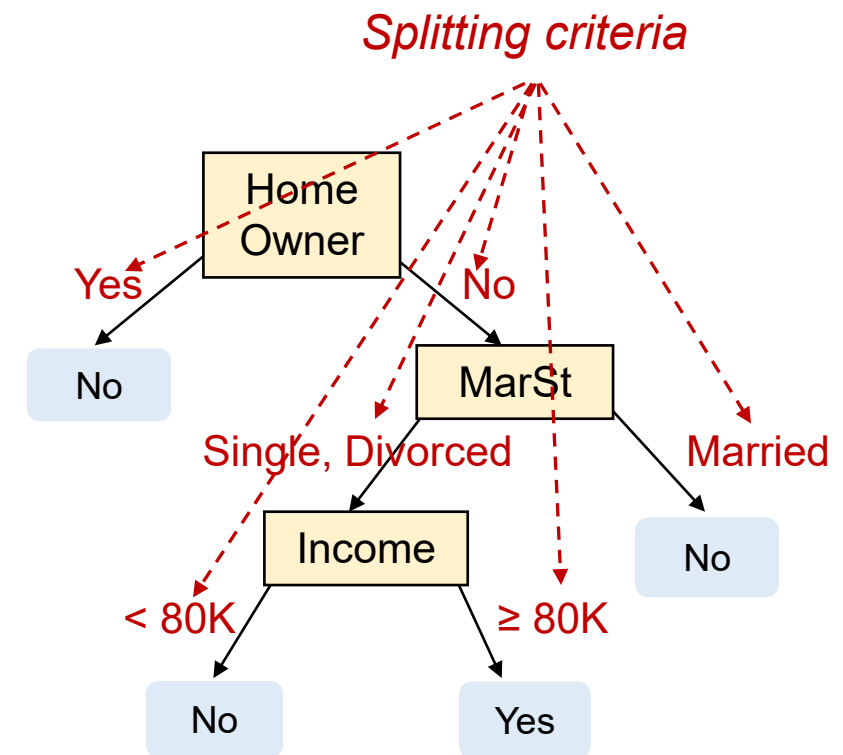


Classification > Methodologies > Decision Tree

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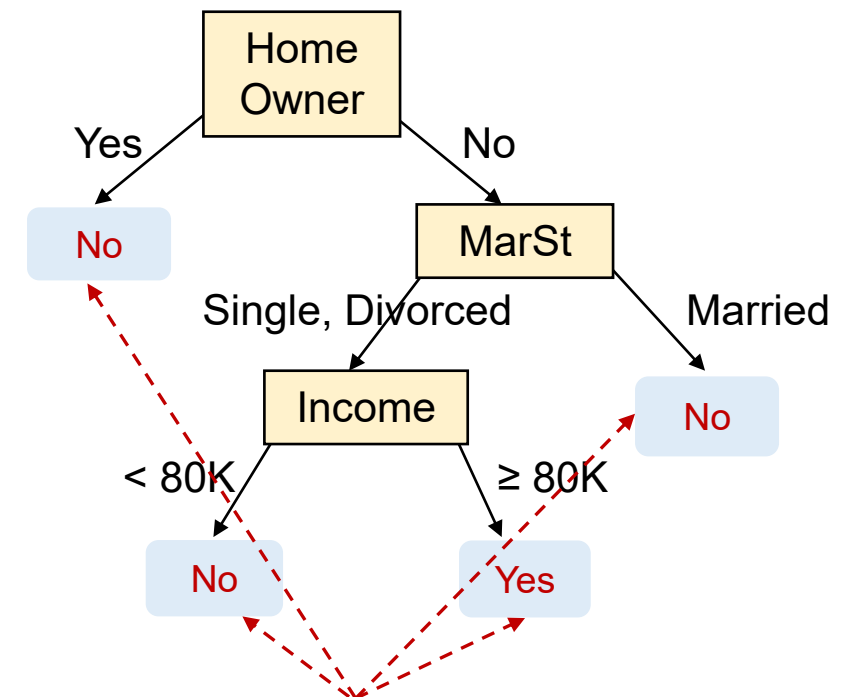


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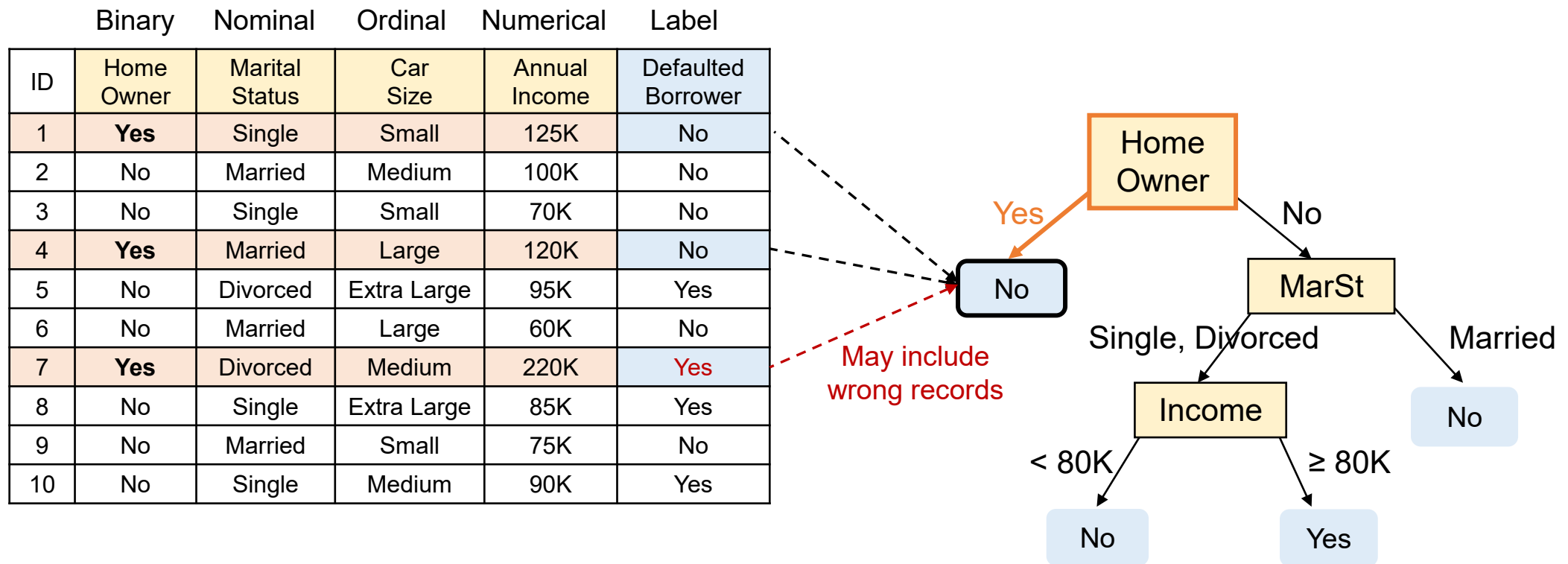
Tree
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→
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a model



Leaf nodes with the final decision on the label (i.e., decision nodes)

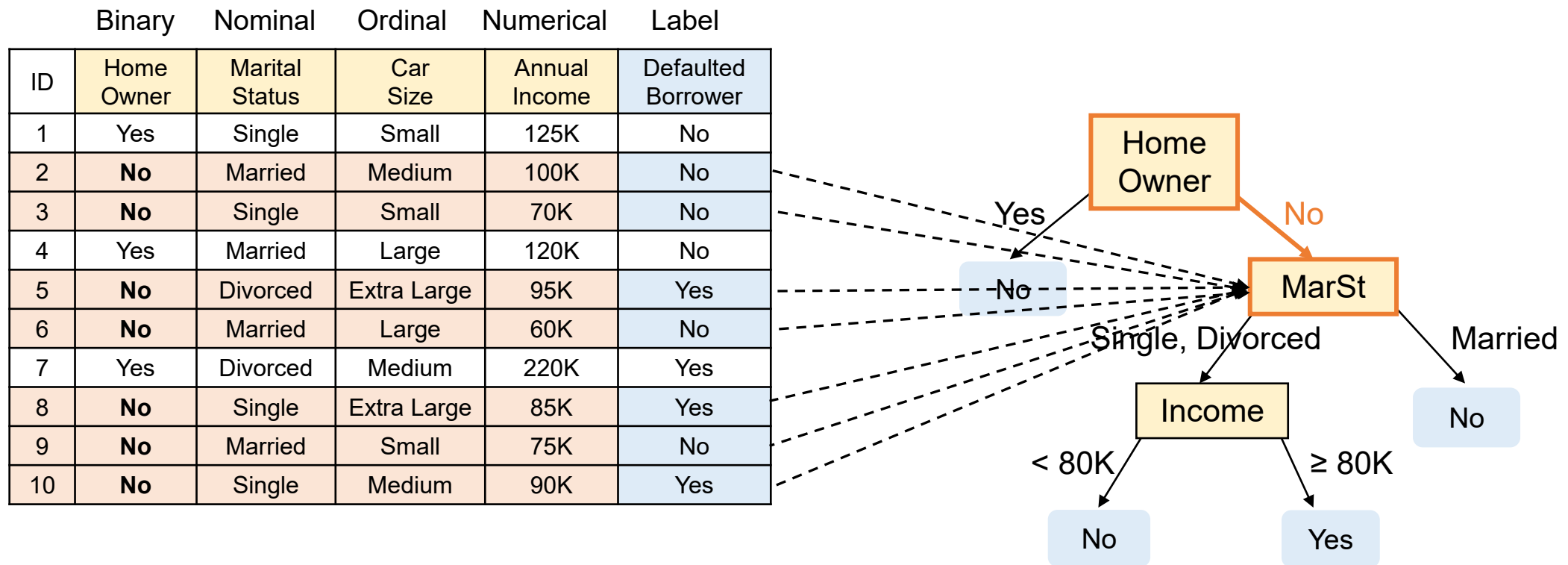
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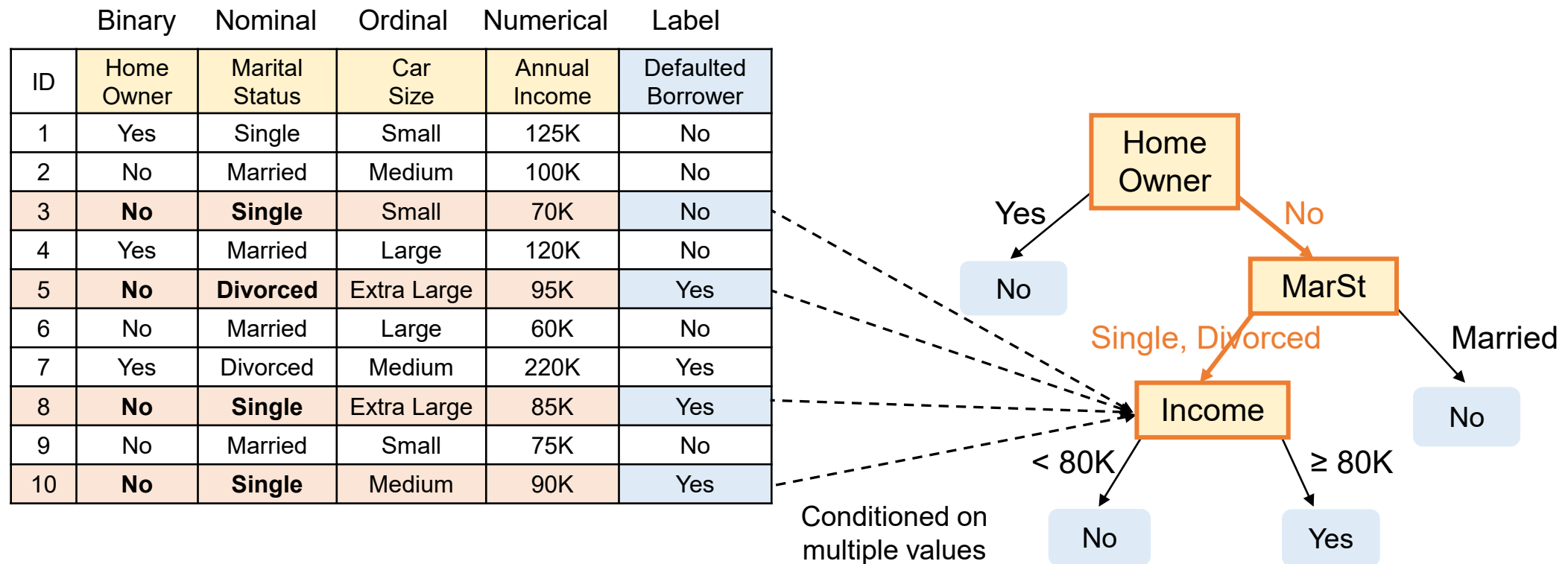
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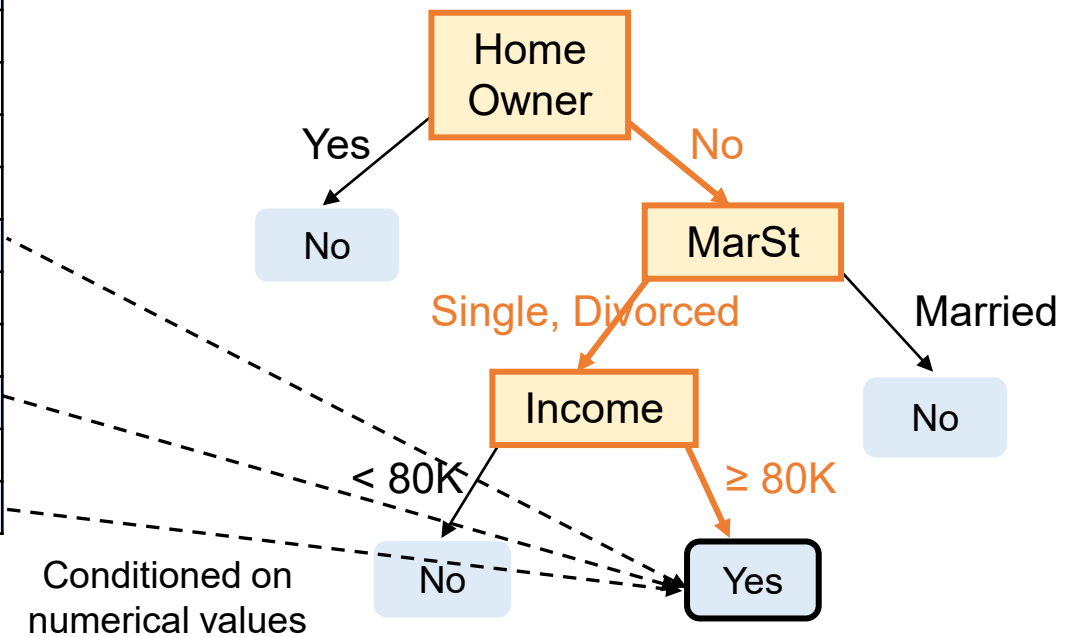
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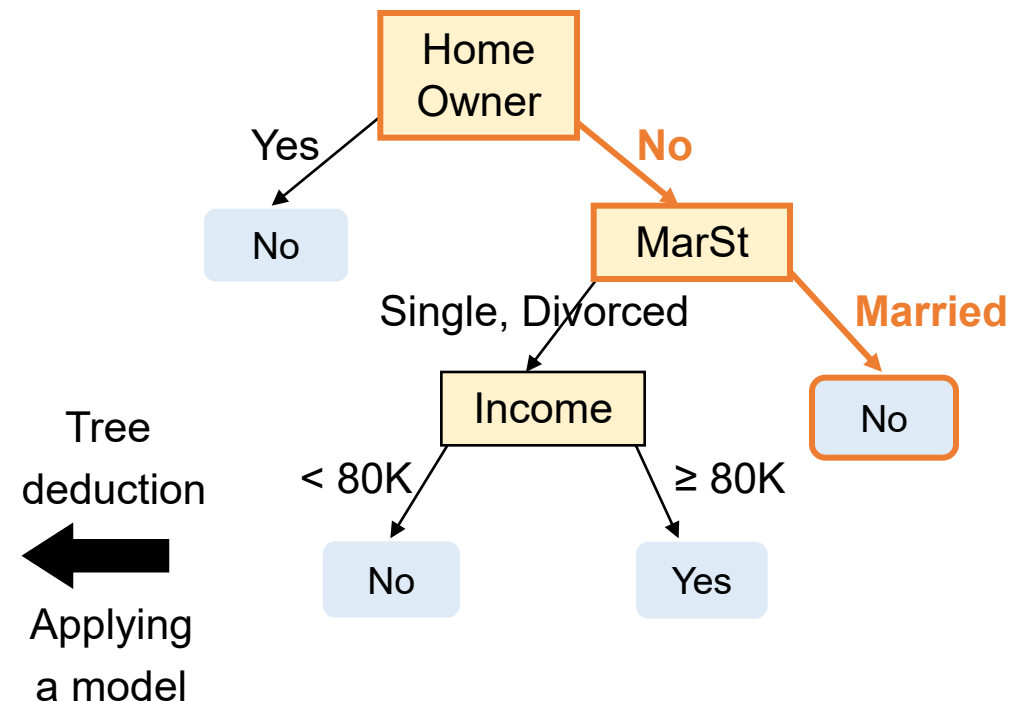
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3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes



Classification > Methodologies > Decision Tree

- A top-down, recursive, divide-and-conquer process to classify data into labels

	Binary	Nominal	Ordinal	Numerical	Label
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	Yes
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes
New	No	Married	Small	80K	No
12	Yes	Married	Large	60K	??
13	No	Single	Extra Large	105K	??



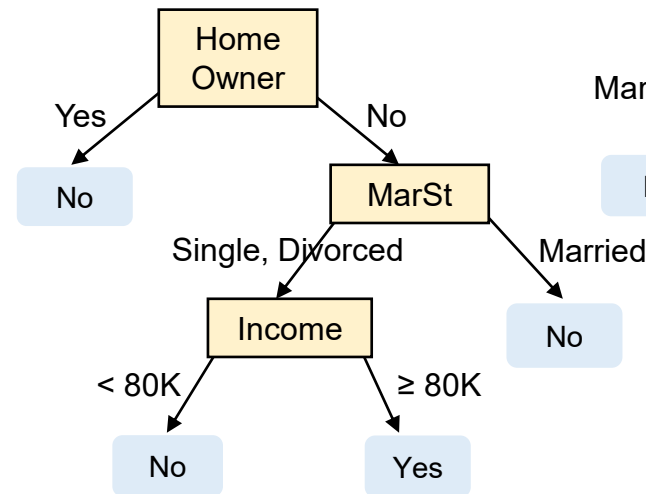
Classification > Methodologies > Decision Tree

- There could be more than one tree that fits the same data
 - *Which tree looks good?*

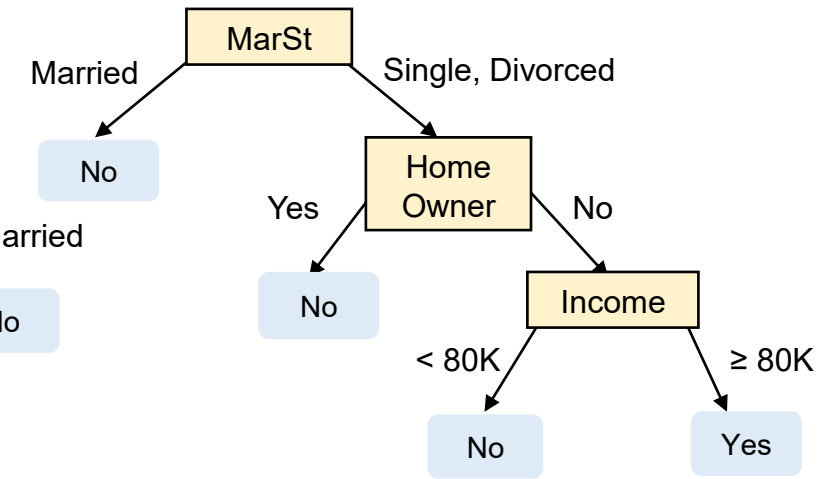
	Binary	Nominal	Ordinal	Numerical	Label
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	Yes
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes



Tree 1



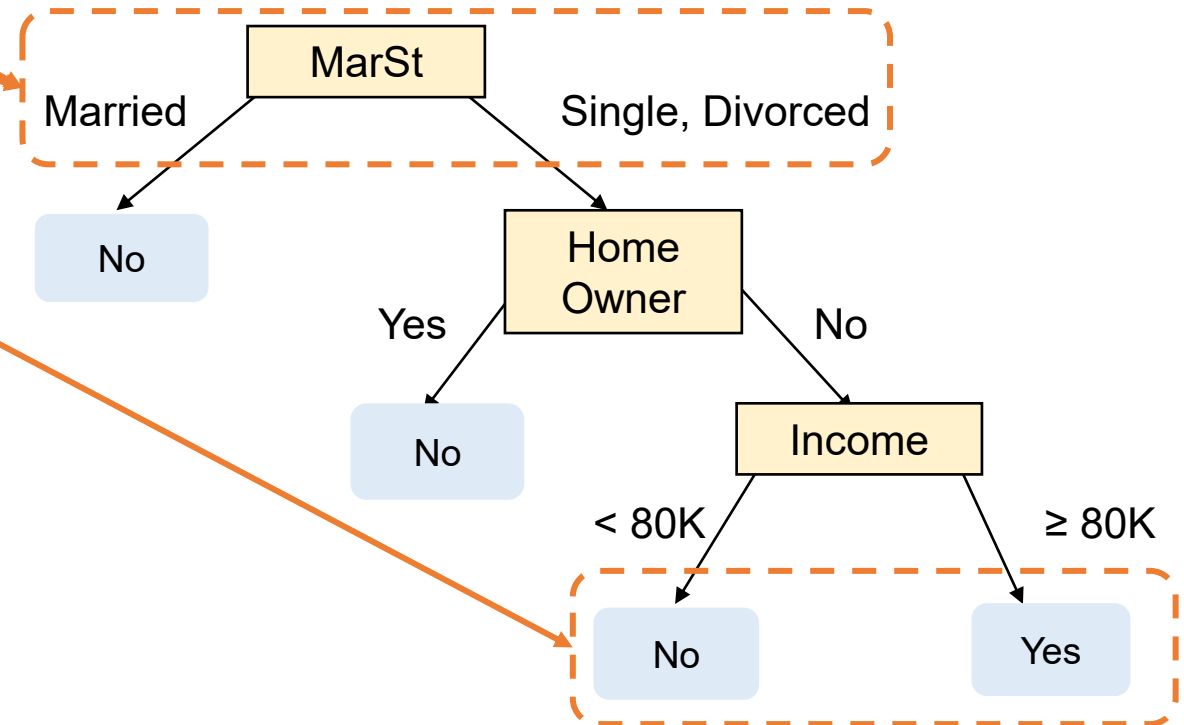
Tree 2



Classification > Methodologies > Decision Tree

□ Two important questions for decision tree induction

- How to split?
 - How to specify the split criterion?
 - How to determine the best split?
- When to stop splitting?

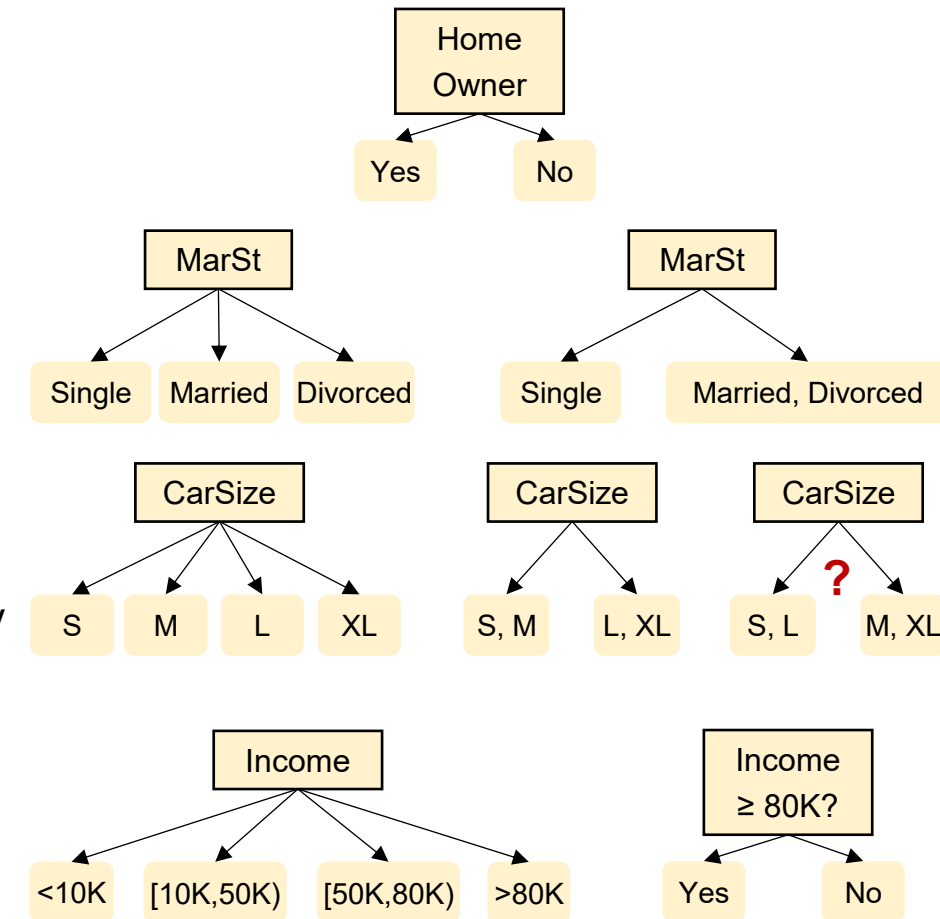


Classification > Methodologies > Decision Tree

□ How to specify the split criterion?

▪ Attribute type

- Binary attribute
 - Binary split
- Nominal attribute
 - Multi-way split: use as many partitions as distinct values
 - Binary split: divides values into two subsets
- Ordinal attribute
 - Multi-way split: use as many partitions as distinct values
 - Binary split: divides into two subsets while preserving order property
- Numerical attribute
 - Multi-way split: discretize to form an ordinal categorical attribute
 - Binary split: consider all possible splits and finds the best cut

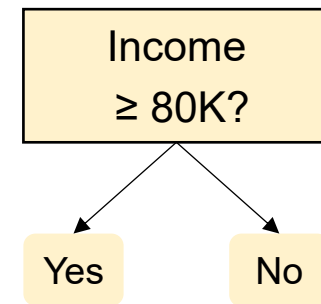
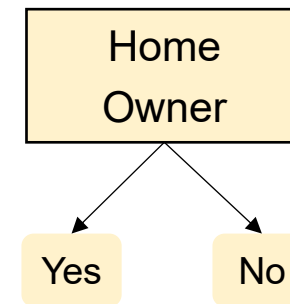
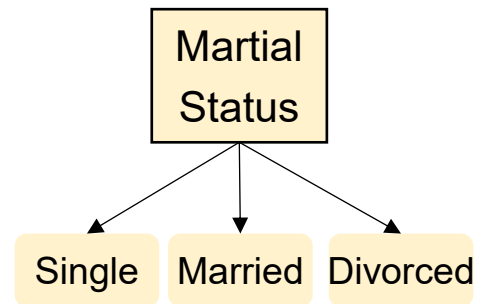


Classification > Methodologies > Decision Tree

□ How to determine the best split?

- Example dataset
 - 5 records of the class label “Yes” and 5 records of “No”
 - Different types of attributes
 - Hone owner, Martial status, Car size, Annual income
- Which split looks good?

	Binary	Nominal	Ordinal	Numerical	Label
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	Yes
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

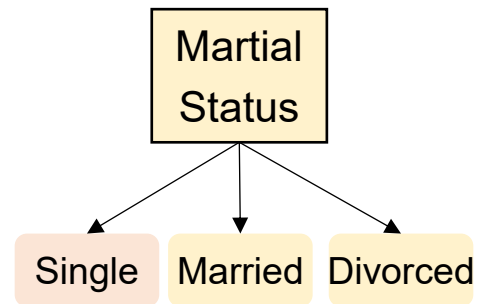


Classification > Methodologies > Decision Tree

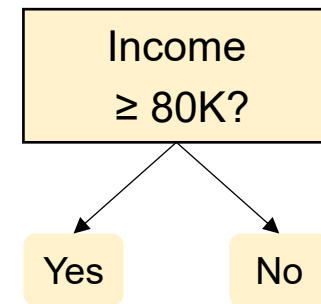
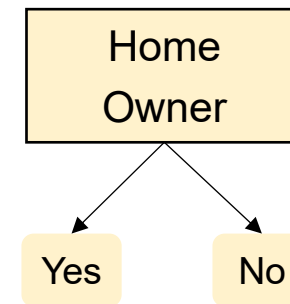
□ How to determine the best split?

- Example dataset
 - 5 records of the class label “Yes” and 5 records of “No”
 - Different types of attributes
 - Home owner, Martial status, Car size, Annual income
- Which split looks good?

	Binary	Nominal	Ordinal	Numerical	Label
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3	No	Single	Small	70K	Yes
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes



of (Defaulted: Yes) 3
 # of (Defaulted: No) 1

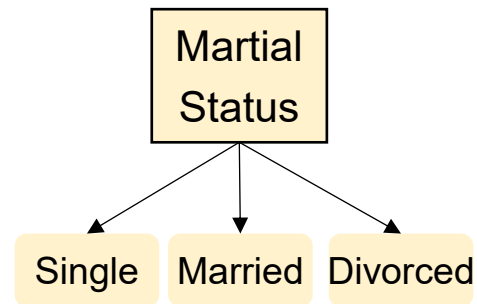


Classification > Methodologies > Decision Tree

□ How to determine the best split?

- Example dataset
 - 5 records of the class label “Yes” and 5 records of “No”
 - Different types of attributes
 - Home owner, Martial status, Car size, Annual income
- Which split looks good?

	Binary	Nominal	Ordinal	Numerical	Label
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	Yes
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

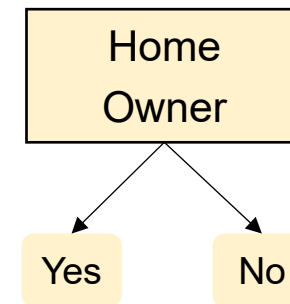


of (Defaulted: Yes)

3 0 2

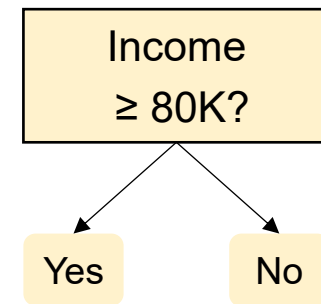
of (Defaulted: No)

1 4 0



1 4

2 3



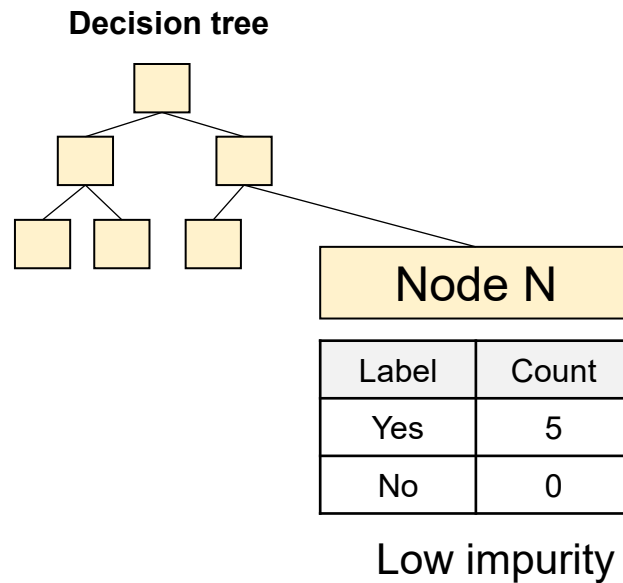
4 1

3 2

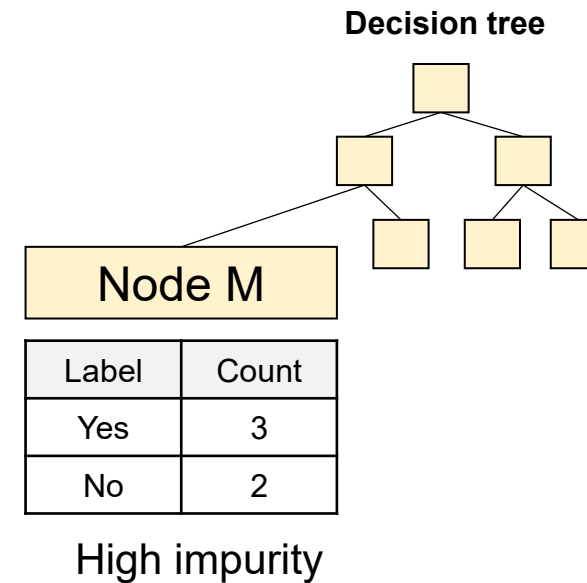
Classification > Methodologies > Decision Tree

□ How to determine the best split?

- Intuition: nodes with purer class distribution are preferred (i.e., low impurity)



VS.



We can say that records falling in node N will have a label “Yes” more likely than those in node M

Classification > Methodologies > Decision Tree

□ How to determine the best split?

- Intuition: nodes with purer class distribution are preferred (i.e., low impurity)
- Greedily seek for higher purity gains
 - Purity measures $f()$ and $g()$

Node N	
Label	Count
C0	N0
C1	N1

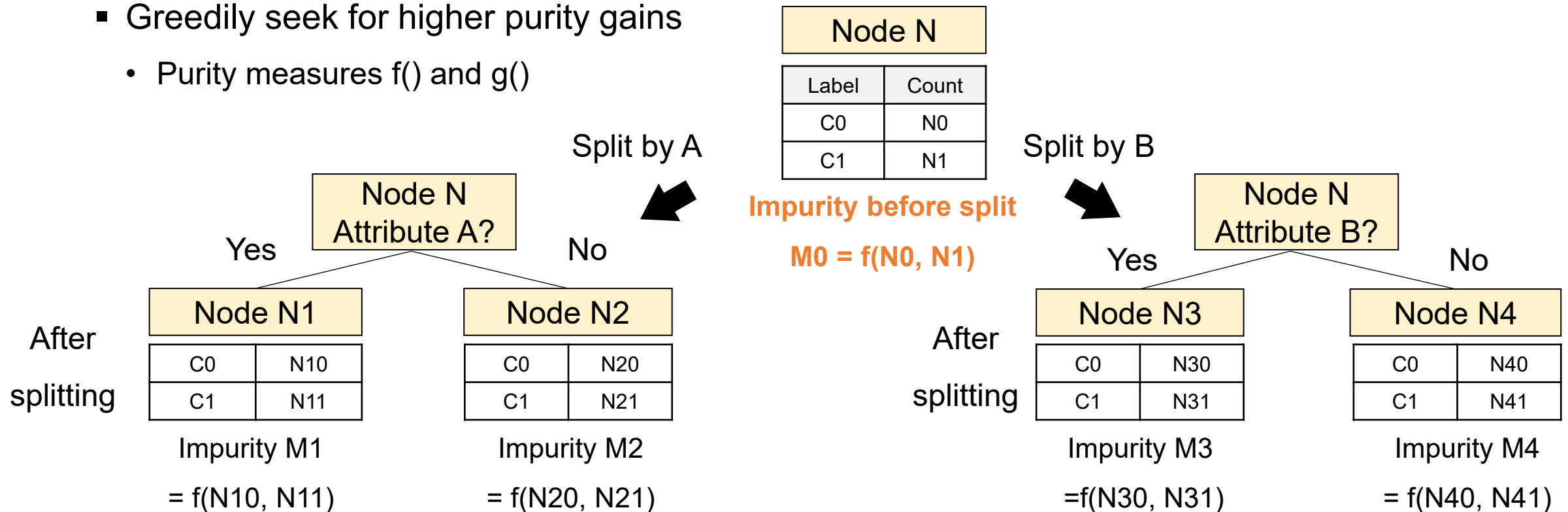
Impurity before split

$$M0 = f(N0, N1)$$

Classification > Methodologies > Decision Tree

□ How to determine the best split?

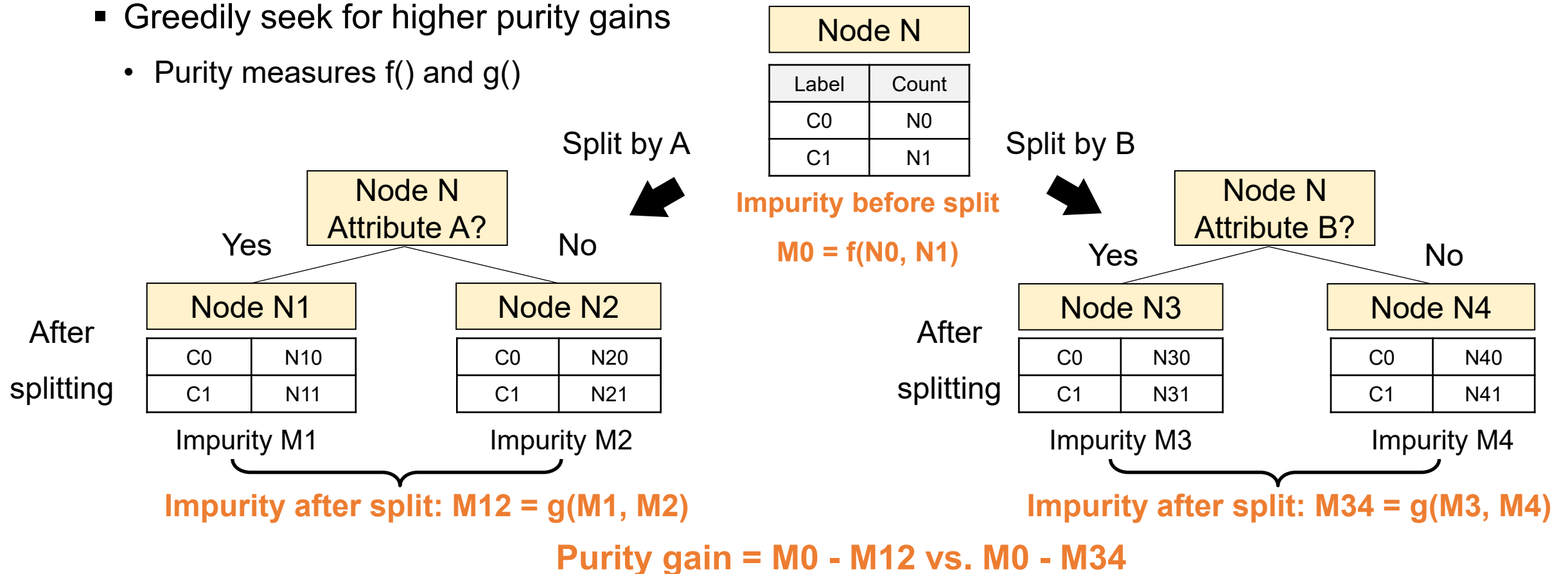
- Intuition: nodes with purer class distribution are preferred (i.e., low impurity)
- Greedily seek for higher purity gains
 - Purity measures $f()$ and $g()$



Classification > Methodologies > Decision Tree

□ How to determine the best split?

- Intuition: nodes with purer class distribution are preferred (i.e., low impurity)
- Greedily seek for higher purity gains
 - Purity measures $f()$ and $g()$



■ Classification > Methodologies > Decision Tree

□ Measures for purity gain in decision tree

- GINI index → CART
- Entropy → ID3
- Gain ratio → C4.5

Outline

- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - **Decision tree**
 - Overview
 - **GINI index** → **CART**
 - Entropy → ID3
 - Gain ratio → C4.5
- ❑ Classification evaluation
- ❑ Ensemble classifiers

■ Classification > Methodologies > Decision Tree > GINI

□ GINI index of a node (i.e., node impurity measure)

- Given a node t , the relative frequency (or probability) $p(j|t)$ of label class j at t

$$p(j|t) = \frac{\text{\# of records of class } j \text{ in } t}{\text{\# of records in } t}$$

- GINI index measures the likelihood of wrong classification, implying node impurity

$$\begin{aligned} GINI(t) &= \sum_j \left(p(j|t) \sum_{k \neq j} p(k|t) \right) = \sum_j p(j|t)(1 - p(j|t)) = 1 - \sum_j p(j|t)^2 \\ &= 1 - \sum_j \left(\frac{\text{\# of records of class } j \text{ in } t}{\text{\# of records in } t} \right)^2 \end{aligned}$$

Classification > Methodologies > Decision Tree > GINI

❑ GINI index of a node (i.e., node impurity measure)

- Given a node t , the relative frequency (or probability) $p(j|t)$ of label class j at t

$$p(j|t) = \frac{\text{\# of records of class } j \text{ in } t}{\text{\# of records in } t}$$

- GINI index measures the likelihood of wrong classification, implying node impurity

$$GINI(t) = \sum_j \left(p(j|t) \sum_{k \neq j} p(k|t) \right) = \sum_j p(j|t)(1 - p(j|t)) = 1 - \sum_j p(j|t)^2$$

$$= 1 - \sum_j \left(\frac{\text{\# of records of class } j \text{ in } t}{\text{\# of records in } t} \right)^2$$

- Example

Node N	
C0	1
C1	5

$$p(C0|N) = 1/6 \quad p(C1|N) = 5/6$$

$$\text{Miss classify C0 as C1} = 1/6 * 5/6 = 5/36$$

$$\text{Misclassify C1 as C0} = 5/6 * 1/6 = 5/36$$

$$GINI(N) = 5/36 + 5/36 = 10/36 = 0.278$$

$$\begin{aligned} GINI(N) &= 1 - p(C0|N)^2 - p(C1|N)^2 \\ &= 1 - (1/6)^2 - (5/6)^2 \\ &= 0.278 \end{aligned}$$

Classification > Methodologies > Decision Tree > GINI

GINI index of a node (i.e., node impurity measure)

- Given a node t and $p(j|t)$ of label class j at t ,

$$GINI(t) = 1 - \sum_j p(j|t)^2 = 1 - \sum_j \left(\frac{\# \text{ of records of class } j \text{ in } t}{\# \text{ of records in } t} \right)^2$$

- For a node t with n records and c classes,

$$0 \leq GINI(t) \leq 1 - \frac{1}{c}$$

- Upper bound (maximum)

$$GINI(t) = 1 - \left(\frac{n/c}{n}\right)^2 - \left(\frac{n/c}{n}\right)^2 \dots - \left(\frac{n/c}{n}\right)^2 = 1 - c \left(\frac{1}{c}\right)^2 = 1 - \frac{1}{c}$$

- When records are equally distributed among all classes, implying high classification error

- Lower bound (minimum)

$$GINI(t) = 1 - \left(\frac{n}{n}\right)^2 - \left(\frac{0}{n}\right)^2 \dots - \left(\frac{0}{n}\right)^2 = 1 - \left(\frac{n}{n}\right)^2 = 0$$

- When all records belong to one class, implying no classification error

Node N	
C0	1
C1	5

$$p(C0|N) = 1/6 \quad p(C1|N) = 5/6$$

$$GINI(N) = 1 - p(C0|N)^2 - p(C1|N)^2 \\ = 1 - (1/6)^2 - (5/6)^2 = 0.278$$

Node N	
C0	3
C1	3

$$p(C0|N) = 3/6 = 0.5 \quad p(C1|N) = 3/6 = 0.5$$

$$GINI(N) = 1 - p(C0|N)^2 - p(C1|N)^2 = 1 - 0.25 - 0.25 = 0.5$$

Node N	
C0	0
C1	6

$$p(C0|N) = 0/6 = 0 \quad p(C1|N) = 6/6 = 1$$

$$GINI(N) = 1 - p(C0|N)^2 - p(C1|N)^2 = 1 - 0 - 1 = 0$$

Classification > Methodologies > Decision Tree > GINI

❑ GINI index example

A set of records

ID	A	B	Label
1	a1	Yes	No
2	a2	No	Yes
...	...	...	...
19	a1	Yes	No
20	a3	No	Yes

Node N		
Before splitting	Label: Yes	9
	Label: No	11

$$GINI(t) = 1 - \sum_j p(j|t)^2$$

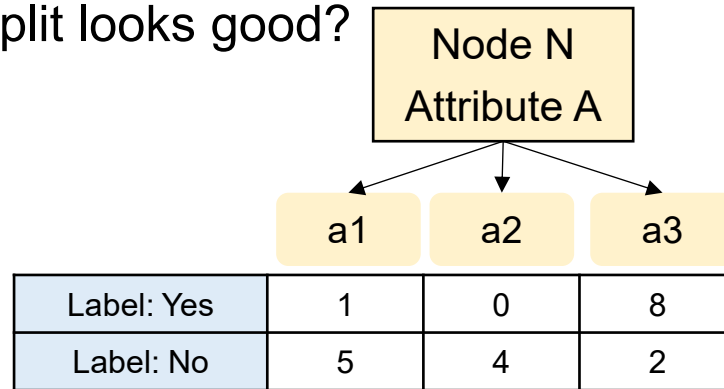
$$p(\text{Label}=\text{Yes}|\text{N}) = 9/20 \quad , \quad p(\text{Label}=\text{No}|\text{N}) = 11/20$$

$$GINI(\text{N}) = 1 - (9/20)^2 - (11/20)^2 = 0.495$$

Classification > Methodologies > Decision Tree > GINI

GINI index example

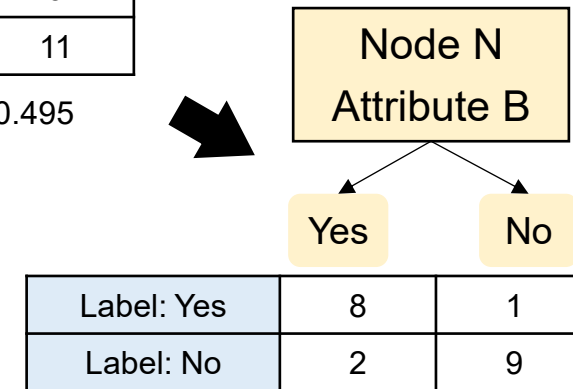
- Which split looks good?



Before
splitting

Node N	
Label: Yes	9
Label: No	11
GINI(N) = 0.495	

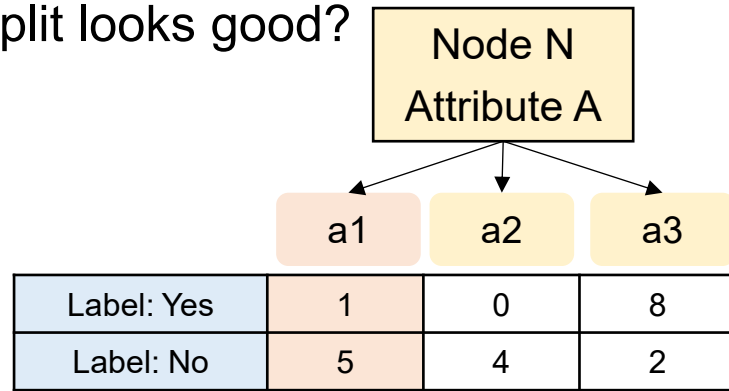
$$GINI(t) = 1 - \sum_j p(j|t)^2$$



Classification > Methodologies > Decision Tree > GINI

GINI index example

- Which split looks good?



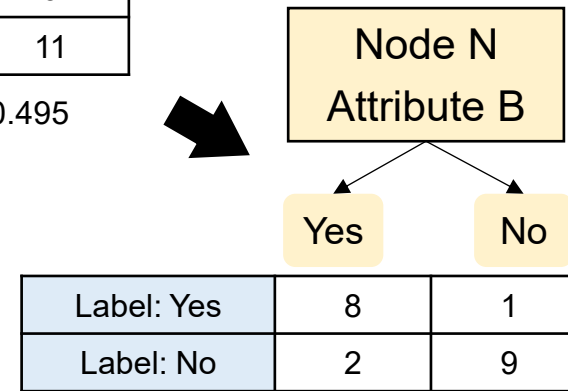
$$p(\text{Label}=\text{Yes}|A=a1) = 1/6, \quad p(\text{Label}=\text{No}|A=a1) = 5/6$$

$$\text{GINI}(A=a1) = 1 - (1/6)^2 - (5/6)^2 = 0.28$$

Before
splitting

Node N	
Label: Yes	9
Label: No	11
GINI(N) = 0.495	

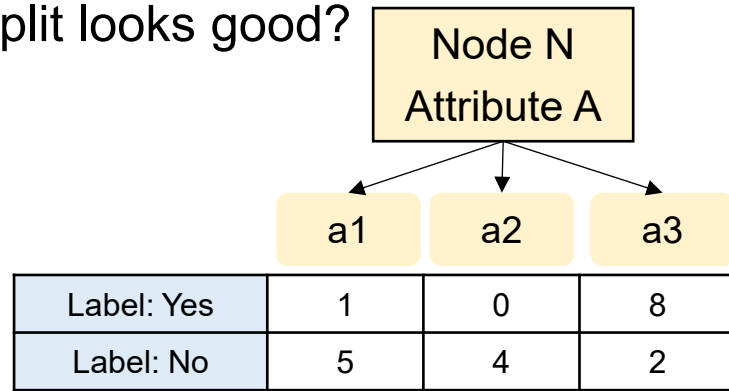
$$\text{GINI}(t) = 1 - \sum_j p(j|t)^2$$



Classification > Methodologies > Decision Tree > GINI

GINI index example

- Which split looks good?



$$p(\text{Label}=\text{Yes}|A=a1) = 1/6, \quad p(\text{Label}=\text{No}|A=a1) = 5/6$$

$$\text{GINI}(A=a1) = 1 - (1/6)^2 - (5/6)^2 = 0.28$$

$$p(\text{Label}=\text{Yes}|A=a2) = \quad, \quad p(\text{Label}=\text{No}|A=a2) =$$

$$\text{GINI}(A=a2) =$$

$$p(\text{Label}=\text{Yes}|A=a3) = \quad, \quad p(\text{Label}=\text{No}|A=a3) =$$

$$\text{GINI}(A=a3) =$$

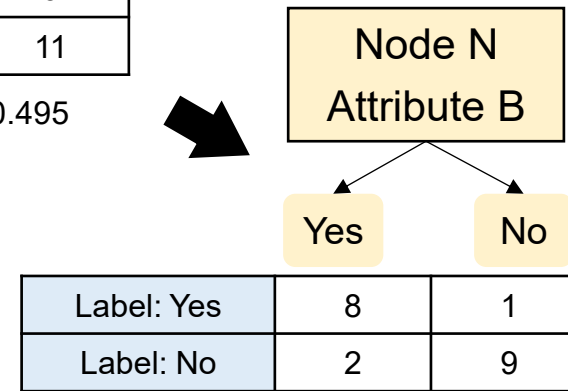
$$\text{Total GINI} = \text{GINI}(A=a1) + \text{GINI}(A=a2) + \text{GINI}(A=a3) =$$

Before
splitting

Node N	
Label: Yes	9
Label: No	11

GINI(N) = 0.495

$$\text{GINI}(t) = 1 - \sum_j p(j|t)^2$$



$$p(\text{Label}=\text{Yes}|B=\text{Yes}) = \quad, \quad p(\text{Label}=\text{No}|B=\text{Yes}) =$$

$$\text{GINI}(B=\text{Yes}) =$$

$$p(\text{Label}=\text{Yes}|B=\text{No}) = \quad, \quad p(\text{Label}=\text{No}|B=\text{No}) =$$

$$\text{GINI}(B=\text{No}) =$$

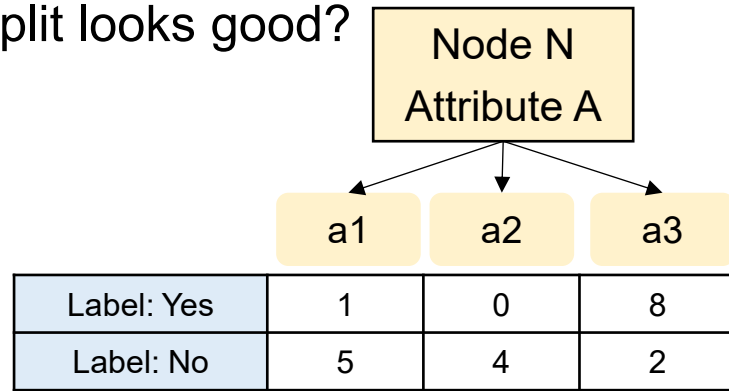
$$\text{Total GINI} = \text{GINI}(B=\text{Yes}) + \text{GINI}(B=\text{No}) =$$

Quiz: which one has “a lower total GINI?” →

Classification > Methodologies > Decision Tree > GINI

GINI index example

- Which split looks good?



$$p(\text{Label}=\text{Yes}|\text{A}=\text{a1}) = 1/6, \quad p(\text{Label}=\text{No}|\text{A}=\text{a1}) = 5/6$$

$$\text{GINI}(\text{A}=\text{a1}) = 1 - (1/6)^2 - (5/6)^2 = 0.28$$

$$p(\text{Label}=\text{Yes}|\text{A}=\text{a2}) = 0/4, \quad p(\text{Label}=\text{No}|\text{A}=\text{a2}) = 4/4$$

$$\text{GINI}(\text{A}=\text{a2}) = 1 - (0/4)^2 - (4/4)^2 = 0$$

$$p(\text{Label}=\text{Yes}|\text{A}=\text{a3}) = 8/10, \quad p(\text{Label}=\text{No}|\text{A}=\text{a3}) = 2/10$$

$$\text{GINI}(\text{A}=\text{a3}) = 1 - (8/10)^2 - (2/10)^2 = 32/100 = 0.32$$

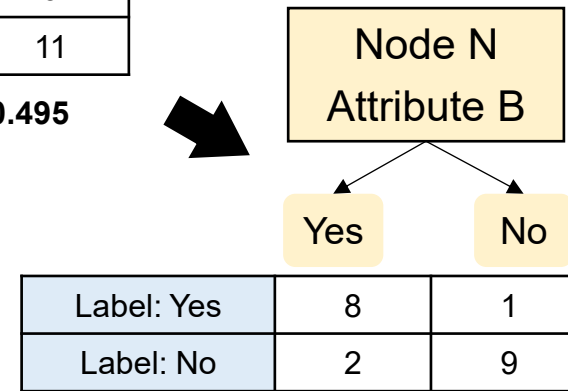
$$\text{Total GINI} = 0.28 + 0 + 0.32 = 0.6$$

Before
splitting

Node N	
Label: Yes	9
Label: No	11

GINI(N) = 0.495

$$\text{GINI}(t) = 1 - \sum_j p(j|t)^2$$



$$p(\text{Label}=\text{Yes}|\text{B}=\text{Yes}) = 8/10, \quad p(\text{Label}=\text{No}|\text{B}=\text{Yes}) = 2/10$$

$$\text{GINI}(\text{B}=\text{Yes}) = 1 - (8/10)^2 - (2/10)^2 = 32/100 = 0.32$$

$$p(\text{Label}=\text{Yes}|\text{B}=\text{No}) = 1/10, \quad p(\text{Label}=\text{No}|\text{B}=\text{No}) = 9/10$$

$$\text{GINI}(\text{B}=\text{No}) = 1 - (1/10)^2 - (9/10)^2 = 18/100 = 0.18$$

$$\text{Total GINI} = 0.32 + 0.18 = 0.5$$

Quiz: which one has “a lower total GINI?” → Attribute B...? but...?

Classification > Methodologies > Decision Tree > GINI

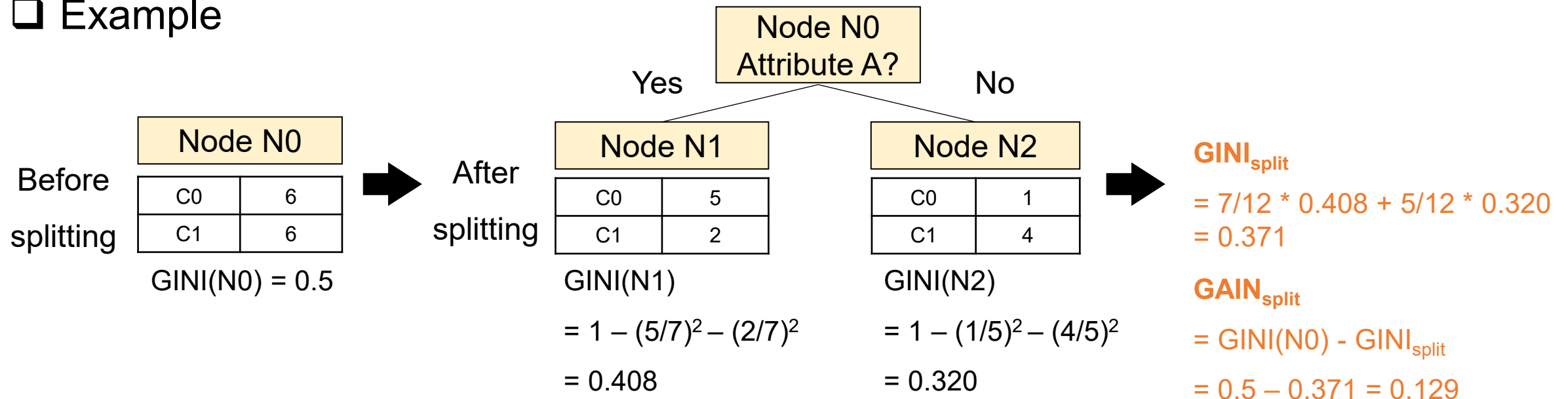
□ When a node t is split into k child nodes, the node impurity of resulting split is

$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i) \quad \begin{array}{l} n = \text{the number of records at node } t \\ n_i = \text{the number of records at } t\text{'s child node } i \end{array}$$

□ The quality of split is computed as the purity gain

$$GAIN_{split} = GINI(t) - GINI_{split}$$

□ Example



Classification > Methodologies > Decision Tree > GINI

❑ Revisit GINI index example

- Which split looks good?

$$GINI(t) = 1 - \sum_j p(j|t)^2$$

$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i)$$

$$GAIN_{split} = GINI(t) - GINI_{split}$$

Attribute A			
	a1	a2	a3
Label: Yes	1	0	8
Label: No	5	4	2

$$GINI(A=a1) = 1 - (1/6)^2 - (5/6)^2 = 0.28$$

$$GINI(A=a2) = 1 - (0/4)^2 - (4/4)^2 = 0$$

$$GINI(A=a3) = 1 - (8/10)^2 - (2/10)^2 = 32/100 = 0.32$$

$$\text{Total GINI} = 0.28 + 0 + 0.32 = 0.6$$

$$GINI_{split} = 6/20 * 0.28 + 4/20 * 0 + 10/20 * 0.32 = 0.244$$

$$GAIN_{split} = 0.495 - 0.244 = 0.251$$

Node N	
Label: Yes	9
Label: No	11

GINI(N) = 0.495

Before
splitting

Attribute B		
	Yes	No
Label: Yes	8	1
Label: No	2	9

$$GINI(B=Yes) = 1 - (8/10)^2 - (2/10)^2 = 32/100 = 0.32$$

$$GINI(B=No) = 1 - (1/10)^2 - (9/10)^2 = 18/100 = 0.18$$

$$\text{Total GINI} = 0.32 + 0.18 = 0.5$$

$$GINI_{split} = 10/20 * 0.32 + 10/20 * 0.18 = 0.25$$

$$GAIN_{split} = 0.495 - 0.25 = 0.245$$

Classification > Methodologies > Decision Tree > GINI

❑ Finding the best split for categorical attributes

- For each distinct value, construct the count matrix
- Choose the split with the least GINI index

Before
splitting

MaritalStatus	
Yes	5
No	5

GINI = 0.5

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	Yes
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

Classification > Methodologies > Decision Tree > GINI

□ Finding the best split for categorical attributes

- For each distinct value, construct the count matrix
- Choose the split with the least GINI index

Before splitting

MaritalStatus	
Yes	5
No	5

GINI = 0.5

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	Yes
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

Multi-way split

	Marital Status		
	Single	Married	Divorced
Yes	3	0	2
No	1	4	0
GINI	0.375	0	0
GINI_{split}	0.15		

Best split

Binary split

	Marital Status	
	Single, Married	Divorced
Yes	3	2
No	5	0
GINI	0.468	0
GINI_{split}	0.374	

	Marital Status	
	Single	Married, Divorced
Yes	3	2
No	1	4
GINI	0.375	0.444
GINI_{split}	0.416	

Classification > Methodologies > Decision Tree > GINI

□ Finding the best split for numerical attributes

- # of possible splitting values?
- Instead of considering every possible split separately,
 1. Sort the records by the target attribute values
 2. Incrementally update the counts and GINI index
 3. Choose the split position that has the least GINI index

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
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7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

Class label (Defaulted)	No		Yes		No		Yes		Yes		Yes		No		No		No		Yes	
Attribute	Annual Income (sorted)																			
	60		70		75		85		90		95		100		120		125		220	
Condition	55																			
	<=	>																		
Yes	0	5																		
No	0	5																		
GINI_{split}	0.5																			

Classification > Methodologies > Decision Tree > GINI

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Class label (Defaulted)	No		Yes		No		Yes		Yes		Yes		No		No		No		Yes	
Attribute	Annual Income (sorted)																			
	60		70		75		85		90		95		100		120		125		220	
Condition	55		65																	
	<=	>	<=	>																
Yes	0	5	0	5																
No	0	5	1	4																
GINI_{split}	0.5		0.444																	

Classification > Methodologies > Decision Tree > GINI

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Class label (Defaulted)	No		Yes		No		Yes		Yes		Yes		No		No		No		Yes	
Attribute	Annual Income (sorted)																			
	60		70		75		85		90		95		100		120		125		220	
Condition	55		65		72															
	<=	>	<=	>	<=	>														
Yes	0	5	0	5	1	4														
No	0	5	1	4	1	4														
GINI_{split}	0.5		0.444		0.5															

Classification > Methodologies > Decision Tree > GINI

□ Finding the best split for numerical attributes

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Class label (Defaulted)	No		Yes		No		Yes		Yes		Yes		No		No		No		Yes			
Attribute	Annual Income (sorted)																					
	60		70		75		85		90		95		100		120		125		220			
Condition	55		65		72		80		87		92		97		110		122		172		225	
	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
Yes	0	5	0	5	1	4	1	4	2	3	3	2	4	1	4	1	4	1	4	1	5	0
No	0	5	1	4	1	4	2	3	2	3	2	3	2	3	3	2	4	1	5	0	5	0
GINI_{split}	0.5		0.444		0.5		0.476		0.5		0.48		0.416		0.476		0.5		0.444		0.5	

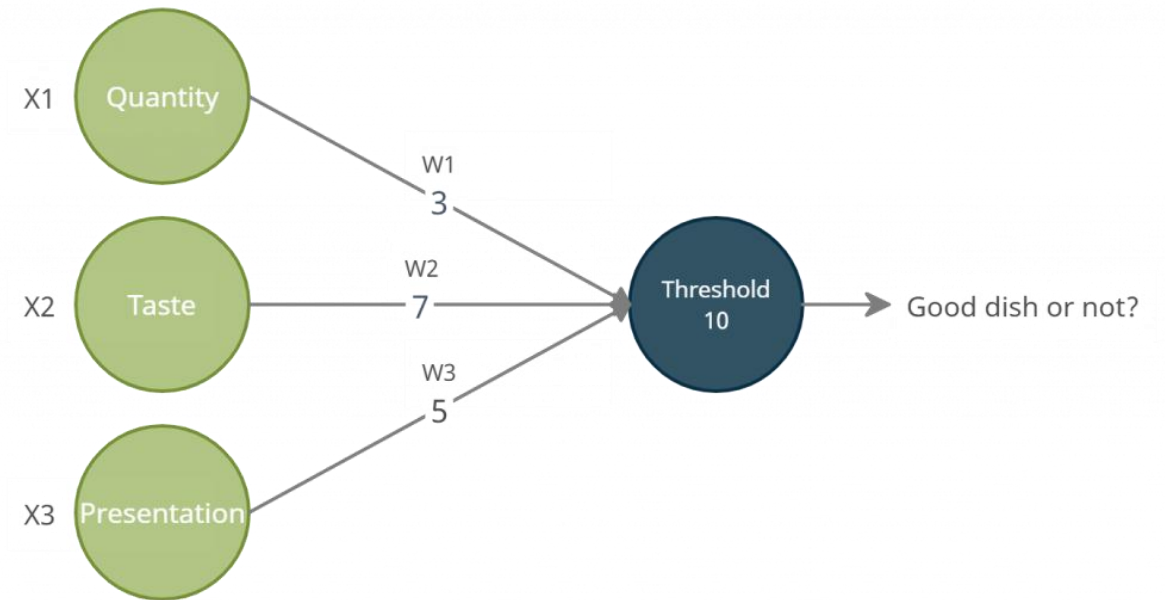
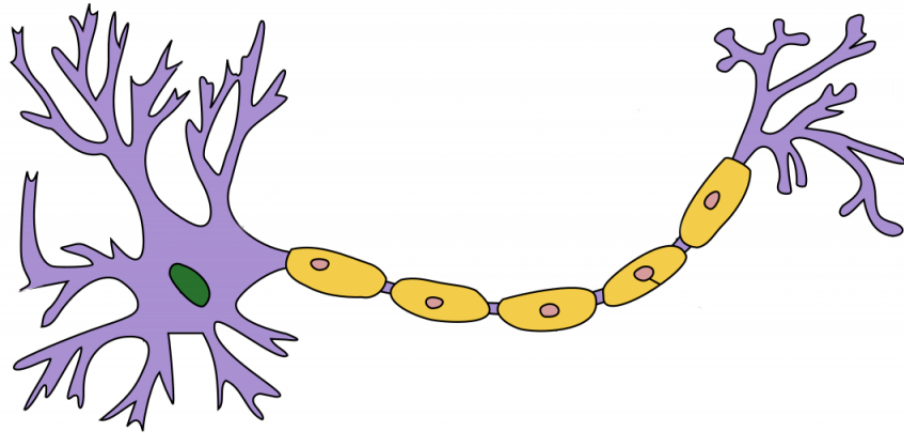
Best split

Next

- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - **Decision tree**
 - Overview
 - GINI index → CART
 - **Entropy** → ID3
 - Gain ratio → C4.5
- ❑ Classification evaluation
- ❑ Ensemble classifiers

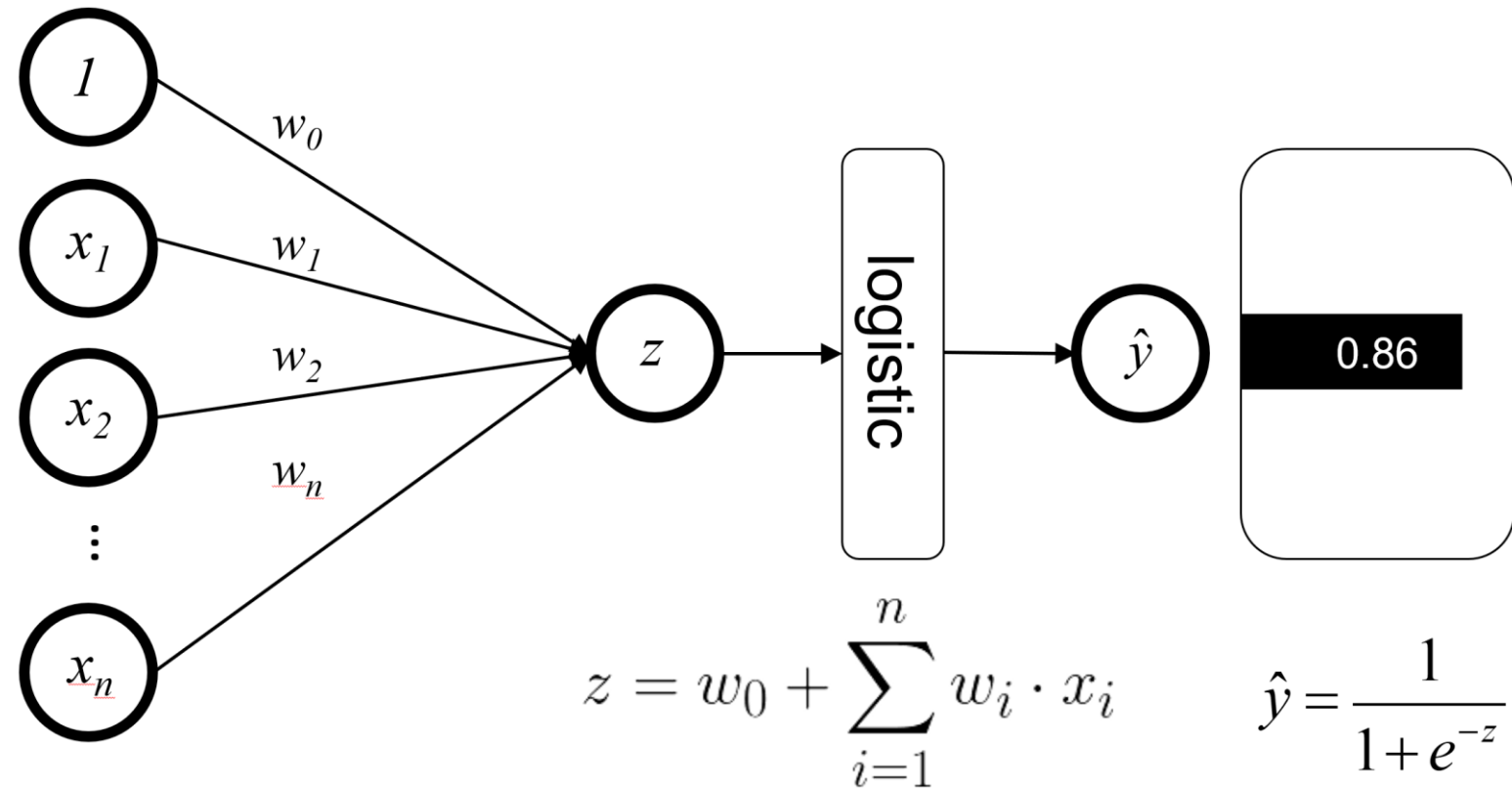
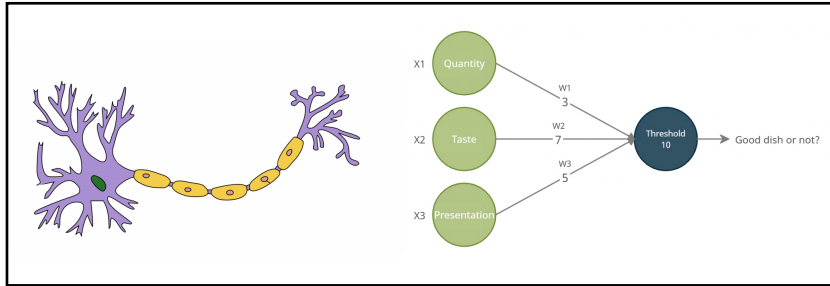
Cookie Slides – Deep Neural Network

- Neuron to perceptron (i.e., artificial neuron)



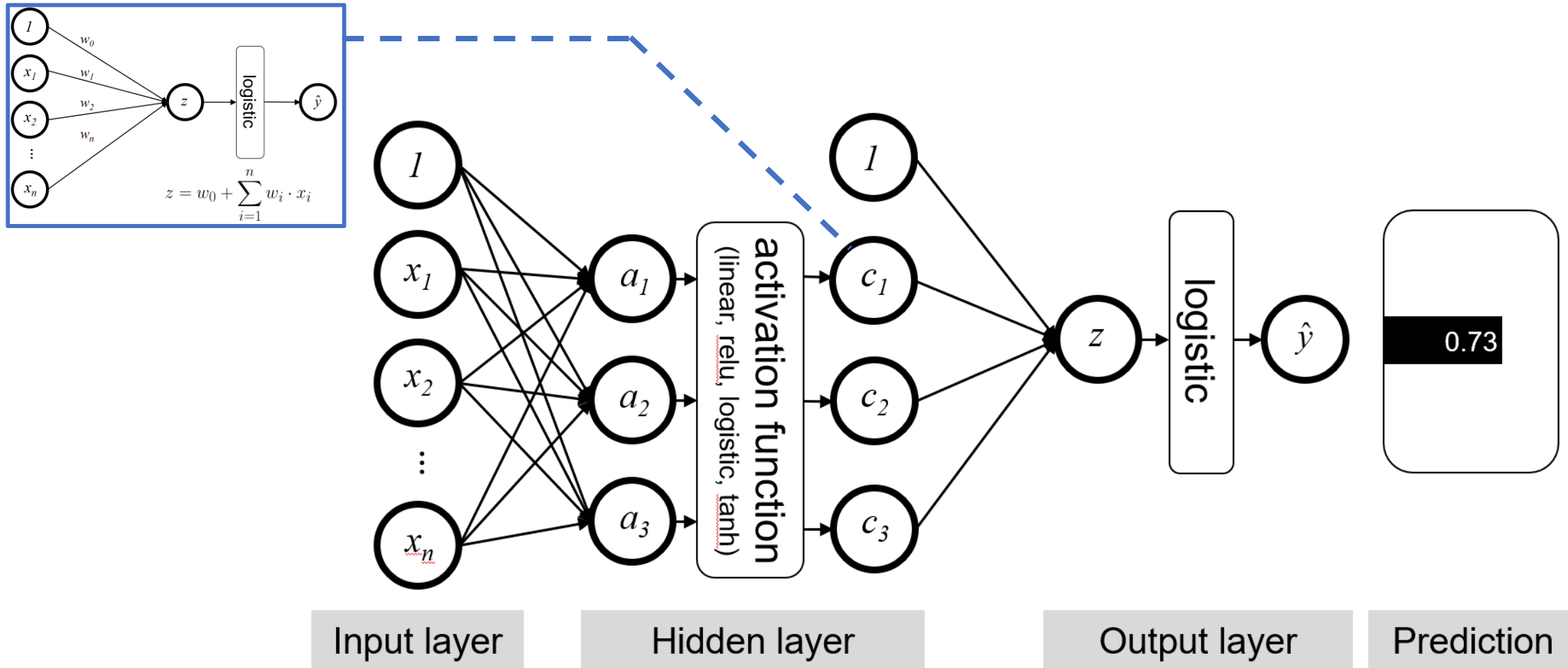
Cookie Slides – Deep Neural Network

□ Logistic regression in the form of perceptron



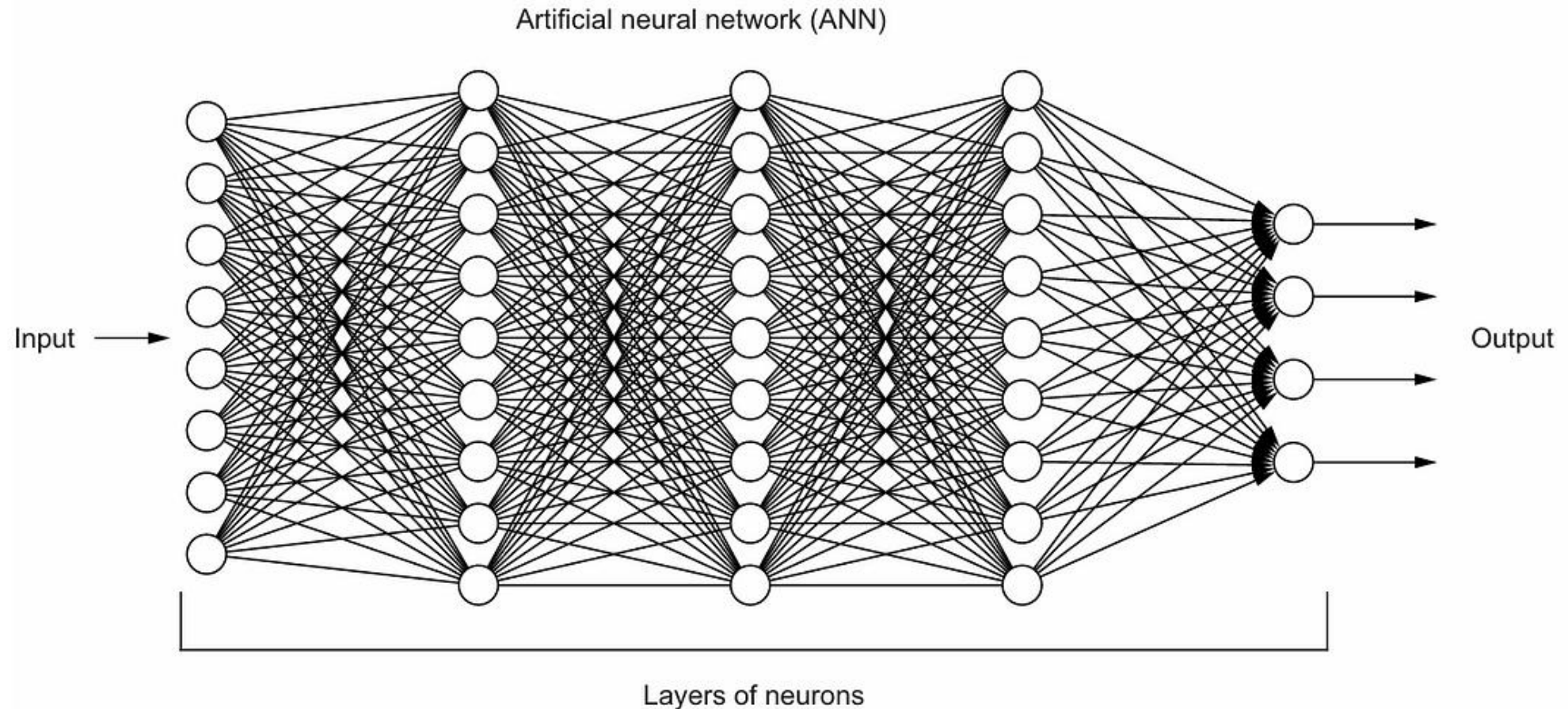
Cookie Slides – Deep Neural Network

- Artificial neural network – a stack of perceptrons with varying activation functions



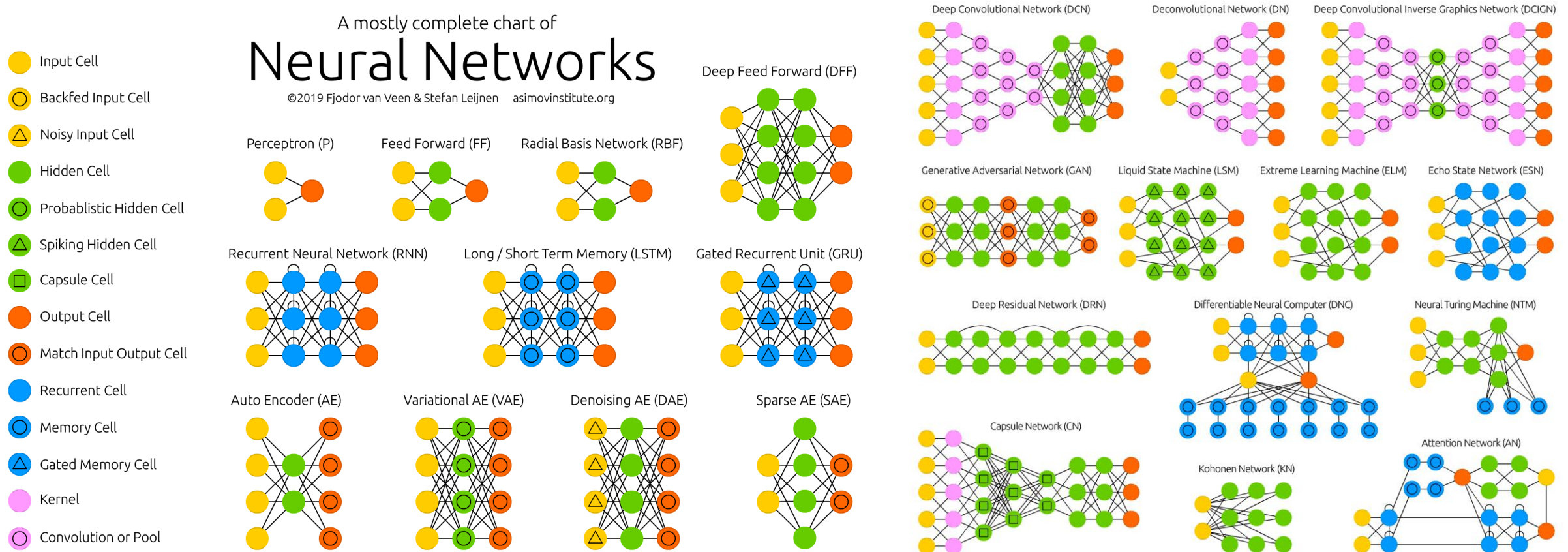
Cookie Slides – Deep Neural Network

- ❑ Deep neural network – a deep stack of perceptrons



Cookie Slides – Deep Neural Network

□ The Neural Network Zoo



Cookie Slides – Eager Learning + Lazy Learning

- ❑ Retrieval-augmented generation (RAG) with k-nearest neighbors
 - Given an **eagerly learned** pre-trained model and a query, it **lazily adapts** with a few relevant examples

